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RESEARCH-ARTICLE

Explaining Recommendations through Conversations: Dialog Model and the Effects of Interface Type and Degree of Interactivity

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Explaining system-generated recommendations based on user reviews can foster users' understanding and assessment of the recommended items and the recommender system (RS) as a whole. While up to now explanations have mostly been static, shown in a single presentation unit, some interactive explanatory approaches have emerged in explainable artificial intelligence (XAI), making it easier for users to examine system decisions and to explore arguments according to their information needs. However, little is known about how interactive interfaces should be conceptualized and designed to meet the explanatory aims of transparency, effectiveness, and trust in RS. Thus, we investigate the potential of interactive, conversational explanations in review-based RS and propose an explanation approach inspired by dialog models and formal argument structures. In particular, we investigate users' perception of two different interface types for presenting explanations, a graphical user interface (GUI)-based dialog consisting of a sequence of explanatory steps, and a chatbot-like natural-language interface.

Since providing explanations by means of natural language conversation is a novel approach, there is a lack of understanding how users would formulate their questions with a corresponding lack of datasets. We thus propose an intent model for explanatory queries and describe the development of ConvEx-DS, a dataset containing intent annotations of 1,806 user questions in the domain of hotels, that can be used to to train intent detection methods as part of the development of conversational agents for explainable RS. We validate the model by measuring user-perceived helpfulness of answers given based on the implemented intent detection. Finally, we report on a user study investigating users' evaluation of the two types of interactive explanations proposed (GUI and chatbot), and to test the effect of varying degrees of interactivity that result in greater or lesser access to explanatory information. By using Structural Equation Modeling, we reveal details on the relationships between the perceived quality of an explanation and the explanatory objectives of transparency, trust, and effectiveness. Our results show that providing interactive options for scrutinizing explanatory arguments has a significant positive influence on the evaluation by users (compared to low interactive alternatives). Results also suggest that user characteristics such as decision-making style may have a significant influence on the evaluation of different types of interactive explanation interfaces.

CCS Concepts: • **Information systems → Recommender systems**; • **Human-centered computing → User studies**; **Natural language interfaces;**

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1 INTRODUCTION

Explaining the decisions or recommendations of artificial intelligence-based systems can yield several advantages for users' perception of these systems. Particularly, presenting explanations in **recommender systems (RS)** can contribute to a positive perception by users in terms of transparency, effectiveness, or trust [107]. In the past, many approaches to explaining the products or services suggested by a RS have been based on ratings assigned by customers with similar preferences, or on features of the recommended items, approaches related to collaborative and content-based filtering methods [45, 110]. More recently, and fueled by the advances in natural language processing, user-written reviews have received considerable attention as rich sources of information about an item's benefits and disadvantages. Here, the wealth of detailed information on the positive and negative aspects of the items can be exploited for generating both recommendations and explanations. However, the question of which review-based information to show and how to present it to impact positively users' perception remains largely open, mainly due a general lack of user-centric evaluations of explanatory RS [90].

Until now, the predominant approach in both RS and **explainable AI (XAI)**, in general, has been to provide explanations in a static manner, i.e., presenting explanations as a single unit of information, whether in numerical, textual or graphical form. The lack of interactivity in these explanation methods limits users' flexibility and control in scrutinizing the reasons of a recommendation according to their individual needs with respect to level of detail, different explanatory aspects, or form of presentation. In this article, we therefore propose and investigate the concept of *interactive, conversational explanations*, where users are free to explore explanatory information in a much less constrained manner. Whereas **conversational RS (CRS)** have become a research topic of interest in recent years (see, e.g., Reference [55]), accessing explanatory information in a conversational manner is a largely unexplored area, with a corresponding lack of empirical evidence for its potential effectiveness [103]. Conversational approaches may be realized in different forms, for example, as natural language-based dialogs with a chatbot, or as interaction steps in a graphical interface. We adopt here the interpretation of the term conversational as used in Reference [55], defining conversational explanations as the provision of explanatory information in an interactive, multi-turn, dialogical process that may be instantiated as natural language dialog, **graphical user interface (GUI)**-based navigation, or other interaction styles.

Our work is grounded on the hypothesis that providing interactive options for examining explanatory information positively impacts users' evaluation of a RS, by allowing the user to request evidence either supporting or contradicting the claims made be the system, that is the recommendations. User reviews are a rich and particularly relevant source for extracting such different types of explanatory information, but translating this information into effective user-system conversations is a largely open question. Consequently, we aim in this article to answer the following research question:

RQ1: How can we leverage conversational explanations to increase perceived explanation quality with respect to explanatory aims such as transparency and effectiveness of the recommendations, and trust in the RS?

A general theoretical model of explainable recommendations describing how explanatory dialogs should be structured has not yet been established. We thus propose, as discussed in Reference [50], to analyze conversational explanations through the lens of argumentation theory, which has produced a wide range of models of argumentation [6]. We follow a dialectical approach [115] with a focus on the exchange of arguments between two parties that provides the basis for dialog models of explanation [77, 114], taking into account the social aspects of an explanatory process [51, 82]. However, the practical application of dialog models in explainable RS and their actual benefit from the users' perspective is yet to be determined.

Founded on the notion of interactive argumentation, we formulated an explanation scheme supporting the exploration of arguments extracted from reviews. We realized this scheme through two types of interfaces: interactive navigation in a GUI and text-based natural language conversation in the form of a chatbot. In the former case, users can navigate through the explanatory information by following links such as "why recommended" or "what was reported" on a certain aspect. In contrast, by using the natural language conversation interface, users can type questions using their own words, within an interaction with a conversational agent. We compared users' evaluation of the RS when interacting with these two explanation interfaces, aiming to answer the following question:

RQ2: How do users evaluate review-based RS, when provided with interactive explanations through different interface types, namely, *GUI-based interactive navigation* and *natural language conversation*?

Despite a considerable body of work on conversational agents, little is still known about how a conversational agent should be conceptualized and designed in the context of XAI, and in particular, in explainable RS. Specifically, there is a lack of knowledge about the type of explanation-related questions users would ask the RS, with a corresponding lack of datasets for training explanation-related intent detection methods. We thus performed a series of studies to build up an annotated dataset of user questions in the domain of hotel recommendations, a domain that represents an interesting mix of search (relevant features are known in advance) and experience aspects (which cannot be fully known until purchase or use [89]) that will likely benefit from third-party opinions [60, 89], expressed in user reviews.

We initially collected user questions in Wizard of Oz pre-study [58] (Section 6.1), which led us to formulating an intent model, which we describe in detail in Section 6.1.3. To assess the extent to which the model is able to accurately represent user intents given a larger set of questions, we developed a first version of a conversational RS and performed a larger corpus collection 6.2. To validate our explanatory intent model, we let users assess the helpfulness of the system's responses, assuming that users will rate the helpfulness of a system response higher if the system has adequately recognized the user's explanatory need. As a first major contribution, our studies resulted in a dataset (ConvEx-DS) in which 1,806 user-generated questions about recommendations are annotated with different types of explanation-related user intents based on the intent model we developed.

As a second major contribution, we developed ConvEx, a conversational agent that includes a **natural language understanding (NLU)** module, based on classifiers trained on ConvEx-DS, using the state-of-the-art **natural language processing (NLP)** model BERT [30], and a dialog policy designed to enable conversation flows given different types of recognized intents. ConvEx allows not only to answer standalone questions but also to provide follow-up arguments to support the system's responses to user requests.

Finally, we conducted a user study to investigate users' evaluation of a RS providing access to explanatory information with two types of interfaces (GUI-based navigation and natural language conversation), and with degrees of interactivity. We also investigated the effects of psychological user characteristics, such as decision-making style and visualization familiarity. Statistical analyses based on Structural Equation Modeling revealed details on the relationships between the perceived quality of explanation and the explanatory objectives of transparency, trust and effectiveness, as well as the influence of user characteristics on these variables. Furthermore, based on work reported by Reference [43], we investigated the impact of our explanation approach on specific aspects of transparency (i.e., information provision: "system provided information to understand how/why is an item recommended"; understanding input: "what data does the system use"; understanding output: "why and how well does an item fit one's preferences"; and interaction: "what needs to be changed for a different prediction," as well as their relation with trust and effectiveness.

In this article, we extend work previously reported in Reference [50] (GUI navigation, referred to in Section 5), in Reference [48] (WoOz pre-study, reported in Section 6.1 and 6.5), and in Reference [49] (dataset collection and annotation, addressed in Section 6.2 to 6.5. New content of this article describes the design and implementation details of ConvEx, a conversational agent to provide natural language explanations (Section 6.6), and the final user study to test effects of different degrees of interactivity and of both types of interfaces: GUI navigation and natural language conversation (Section 7).

The contributions of this article can be summarized as follows:

- We introduce a novel concept of conversational explanations in RS based on a scheme for explanations as interactive argumentation.
- We provide an intent model for natural language explanatory queries, categorizing intents along different dimensions and validate it in the domain of hotel recommendations.
- We release the dataset **ConvEx-DS**[1] consisting of 1806 user questions with explanatory purpose in the domain of hotels, with question intent annotations, facilitating the development of explanatory conversations in RS.
- We present **ConvEx**, a novel conversational agent for explainable RS, which supports conversation flows for different types of intents, providing explanations extracted from user reviews.
- We provide empirical evidence of the positive effect of interactive explanatory interfaces (both GUI navigation and natural language conversation) on users' evaluation of RS.
- Applying Structural Equation Modeling, we reveal details of the relationships between perceived explanation quality, transparency, recommendation effectiveness, and trust, as well as between specific aspects regarding perceived transparency.

The following sections are organized as follows: We first discuss existing work related to our research questions. Subsequently, as a theoretical framing of our approach, we discuss the notion of explanations as interactive argumentation. Next, we describe the basic technical methods we applied to construct an explainable CRS based on reviews. Sections 5 and 6 present the design and development of the GUI-based and natural language-based CRS, with a stronger focus on the natural language-based system, since the GUI-based system has been described in earlier work [50], while the system ConvEx is a new contribution. Section 7 describes as a main contribution a user study evaluating the effects of interface style and degree of interactivity, which were analyzed through a SEM. Finally, we present our conclusions, limitations of the present work, and future research options.

[1]ConvEx-DS can be downloaded at https://github.com/intsys-ude/Datasets/tree/main/ConvEx-DS.

2 RELATED WORK

2.1 Explanations in RS

Providing explanations for recommended items can serve different purposes. Explanations may enable users to to understand the suitability of the recommendations, to understand why an item was recommended, or they may assist users in their decision making. Providing explanations can also increase users' perception of transparency and their trust in the RS [94, 106]. Among the most popular explanation approaches are the methods based on collaborative filtering (e.g., "your neighbors' ratings for this movie" Herlocker et al. [45]), as well as content-based methods that allow feature-based explanations, showing users how relevant item features match their preferences (e.g., Vig et al. [110]).

Review-based explanations. As summarized in Reference [50], review-based explanatory methods leverage user generated content, rich in detailed evaluations on item features, which cannot be deduced from the general ratings, thus enabling the generation of more detailed explanations, compared to collaborative filtering (e.g., "Your neighbors' ratings for this movie" [45]) and content-based approaches (e.g., Reference [110]). Review-based methods allow to provide: (1) verbal summaries of reviews, using abstractive summarization through **natural language generation (NLG)** techniques [15, 26], (2) a selection of helpful reviews (or excerpts) that might be relevant to the user, detected using deep learning techniques and attention mechanisms [18, 32], and (3) an account of the pros and cons of item features, usually using topic modelling or aspect-based sentiment analysis [31, 119, 123]. This information can also be integrated into RS algorithms like matrix or tensor factorization [5, 116, 123]) to generate both recommendations and aspect-based explanations.

Our work is based on the third approach, as reported in Reference [50], and is particularly related to the Explicit Factor Model proposed by Zhang et al. [123], which integrates content extracted from reviews in a Matrix Factorization method. Based on this model, we can obtain statistical information on users' opinions (which has been proven to be useful for users [46, 87]), that can be provided in explanations through different presentation styles (verbal or visual). Moreover, we go beyond the basic model by examining interaction possibilities to explore the reasons behind the system's assertions at different levels of detail.

2.2 Interactive Explanations

As discussed in Reference [50], effects of interactivity have been studied widely in fields like online shopping and advertising [71, 104], and more specifically in the context of critique-based RS, where users are able to specify preferences and further criteria to refine the recommendations provided (e.g., References [20, 73, 74]). Despite the intuitive advantages that interactivity can bring, it does not always translate into a more positive attitude toward the system, since it also depends on the context and the task performed [71]. Nevertheless, it has been shown that higher active control may be beneficial in situations where users have more information needs and a clearer mental goal [71], which also applies in our case (i.e., deciding which hotel to book).

Limited work in XAI has so far addressed interactive explanations, although to a much lesser extent compared to static explanations [1]. As discussed in Reference [50], the dominant trend has been to provide mechanisms to check the influence that specific features, points, or data segments have on the predictions of **machine learning (ML)** algorithms, as in References [22, 63, 102]. However, the impact of such interactive approaches in explainable RS on the user remains largely unexplored. More specifically, the dominant ML interactive approach differs from ours in at least two ways: (1) we use non-categorical sources of information, subjective in nature and unstructured, which, however, can be used to generate both textual and visual structured arguments, and (2) existing approaches are mainly designed to meet the needs of domain experts, i.e., users with prior knowledge of **artificial intelligence (AI)**, while we target the general public.

2.3 Conversational Recommender Systems (CRS)

While a widely accepted definition of CRS has not yet been established, we adopt the following statement by Jannach et al. [55], "A CRS is a software system that supports its users in achieving recommendation-related goals through a multi-turn dialogue." This definition does not limit the way in which input is made by the user, so the term *conversational* does not exclude forms of interaction outside written or spoken text. Furthermore, and according to Reference [55], input and output modalities of CRS, as found in CRS related literature, involve approaches based on:

(1) Conventional web-based navigation, based on structured layouts and features, like buttons and hyperlinks. Throughout this article, we will refer to this modality as *GUI navigation*. Particularly, we address our proposal of explanatory interface under this paradigm in Section 5.
(2) Natural language (written or spoken).
(3) Hybrid, a combination of natural language and other modalities. Under this approach, users can indicate their input using both natural language expressions and features such as buttons and other web controls. In Section 6, we discuss our approach to explanations under this paradigm, which we will refer to as *natural language conversation* throughout the article.

A variety of CRS have been developed based on written or spoken text (e.g., References [53, 105]), but GUI-based CRS are also widely in use, especially in commercial applications. As summarized by Reference [55], CRS perform four main types of tasks: requesting user's preferences, recommending items, explaining decisions/predictions and responding to users' actions. However, as pointed out by Reference [55], conversational explanations have been addressed to a much lesser extent, compared to the other CRS tasks, with only a few papers reported on this subject.

Our work differs from the existing approaches to CRS, as in work reported by References [23, 122], who aimed to collect user preferences to generate recommendations through dialog in domains such as restaurants or electronics, usually neglecting explanations for recommended items, which is our central purpose. Only very limited work has addressed explaining recommendations in a conversational manner. Chen et al. [21] propose a CRS aimed at explaining recommendations but their explanations are based on item features and do not exploit the rich information in reviews. Also, their evaluation of the quality of generated explanations is based on the use of offline evaluation metrics such as BLEU [91] and ROUGE-L [69], while in our work we assess the system and its explanations in terms of the users' evaluation. Zhao et al. [124] present a method for generating reasons for recommendations in a CRS based on user comments, which is, however, limited to explaining single items returned upon an explicit user query without providing summative or comparative explanations, as is the case in our approach. Our work also differs from conversational approaches in the hotel domain, which focus on processes like customer service and booking assistance [13]. We aim, in contrast, to explore the implications and effects of using conversational interfaces to explain the rationale of the RS. In one of the few works on the subject, the authors of Reference [92] propose a model of social explanations for movie recommendations. However, according to their approach, it is the system that leads the conversation, providing justifications for recommendations even when they are not explicitly requested by the user, whereas according to our proposal, the user would have the active role, being enabled to ask the questions that lead to an argumentation by the system. While this requires users to express their information needs explicitly, which may not be easy if their preferences are not clear, it is also more targeted and may be more effective, possibly also contributing to preference construction in the interaction process.

2.4 Dialog Models of Explanation

As discussed in Reference [50], in contrast to static approaches to explanation, dialog models have been formulated conceptually [2, 77, 97, 112], allowing arguments over initial claims in explanations, and leading to an interactive exchange of statements. For example, Rago et al. [97] defined a protocol to provide conversational argumentative explanations in RS. They restrict, however, user interactions to a limited set of possible questions a user may ask, while we explore possibilities for users to express their explanatory needs in their own words.

This dialogical approach contrasts with other argumentative—though static—explanation approaches [4, 16, 46, 64, 121] based on static schemes of argumentation (e.g., References [40, 108]), where little can be done to indicate to the system that the explanation has not been fully understood or accepted, and that additional information is still required, as initially discussed in Reference [50]. Despite the potential benefit of using these models to increase users' understanding of intelligent systems [82, 117], their practical implementation in RS (and in XAI in general) still lacks sufficient empirical validation [77, 82, 103].

2.5 Question Answering (QA)

Our work is closely related to QA systems, which aim to answer questions posed by users in natural language, using **information retrieval (IR)** and NLP techniques, on various types of web documents or in knowledge bases, as initially discussed in Reference [48]. While most QA systems are designed to reply to questions by offering excerpts from documents or lists of items, much less work has been devoted to advanced "how-to," "why," evaluative, comparative, and opinion questions [68, 83] that usually require the aggregation and comparison of multiple items over different pieces of information. Lipton [70] defines *explanation* as an answer to a *why-question*, however, other types of questions can also be answered by explanations, i.e., how? what? [82], the latter being one that could be answered with a factoid sentence, for which we aim to support both factoid and advanced question types. Additionally, and in contrast to the common QA approach where the system replies to standalone questions, interactive QA involves a dialog interface enabling related, follow-up and clarification questions [96]. To train NLP methods recognizing these different types of user intents, a high-quality dataset of annotated questions is required. While several datasets exist that support the development of **question-answering (QA)** systems, these are generally focused on open domain-search (e.g., References [81, 98]), or on specific processes such as flight or hotel booking, without a focus on explanatory interaction. This gap led us to develop a dataset that is focused on explaining recommendations based on arguments in user reviews.

Our approach, as elaborated in Reference [48], differs from most QA methods, especially those based on IR, because in our case responses are not generated solely on the basis of information sources, but should be consistent and reflect the mechanism used to generate the recommendations. Additionally, to answer complex questions (e.g., "why"), our approach focuses on the aspects most relevant for users, to provide concise and relevant statements, aggregating information from different reviews. To this end, our approach relies on the user profile inferred by the RS algorithm, especially when no explicit features are addressed in the user's question. Implicit user preferences are not taken into account in most QA approaches, which stems from their use of IR methods, where the relevance of a document is estimated based on how much its content is related to the query [84]. Additionally, we propose to follow an argumentative explanation structure to generate responses, which could improve users' evaluation of RS, as evidenced by using an interactive GUI navigation approach, and as we reported in Reference [50]. Although argumentation has already been exploited in QA [86], its use has been mainly focused on the algorithmic aspects of information extraction, whereas very few approaches exploit argumentation as a way of presenting explanations in response to user queries [3].

2.6 Dialog Systems and Conversational Agents

Often referred to as conversational agents, or dialog agents, dialog systems enable human-computer interaction by means of natural language expressions. Dialog systems can support interactions based on speech or written text. Such systems can be categorized as non-task-oriented (e.g., smalltalk systems) or task-oriented (e.g., booking or searching, or general assistants like Siri or Cortana) [19]. Our approach falls into the text-based, task-oriented group. Throughout this article, we will refer to the terms dialog system and conversational agent interchangeably.

In Reference [19], authors discuss two types of methods for implementing task-oriented dialog systems: pipeline and end-to-end methods. The first is the most widely applied, and is characterized by architectures that include the following components: language understanding (NLU modules to interpret intent and slots), a dialog state tracker (establishes the dialog state based on input and dialog history), dialog policy learning (to determine the next action in the dialog), and **natural language response generation (NLG)**. However, end-to-end methods allow for joint training of all components. This approach can be beneficial in contexts where flexible adaptation is required, e.g., when the system is to be rapidly scaled to new domains or applications. However, the interdependency between pipeline components could make it difficult to adapt to new domains or data types (changes in one component may require changes in other components). Our implementation is framed within the pipeline approach, which remains beneficial for the early stages of dialog engineering in new domains [19], as in our case (explainable RS), and we leave for future work the exploration of end-to-end approaches.

Dialog management. The components of dialog state and dialog policy are usually grouped into a **dialog management (DM)** component. A taxonomy of DM approaches is discussed by Harms et al. [42], who differentiate between two main approaches: handcrafted (state and policy are defined as a set of rules defined by developers and domain experts) and probabilistic (rules are learned from corpora with real conversations). Our implemented system ConvEx falls into the former class, given that, although we compiled a dataset useful to detect user intention (ConvEx-DS), we still lack a corpus to facilitate the inference of states and sequences for conversations in the explanatory RS context.

Particularly, our implementation falls into the handcrafted approach known as finite-state, where the dialog state has a fixed set of possible transitions to other states. In our implementation, this was parameterized in a static file, where we set the sequence of possible actions to be executed by the system, given the recognized intent for a user query (see Section 6.6). More sophisticated approaches such as frame-based ones are used in commercial dialog management systems like Google's DialogFlow,[2] which allows greater flexibility, e.g., adding a data model so that slots can be filled in any dialog sequence, which was not necessary for the purposes of this article.

2.7 Moderation Effect of User Characteristics on Users' Evaluation of RS and Explanations

Individual user characteristics such as personality, domain knowledge, or trust propensity may impact users' assessment of a CRS and trust in the system [14]. It has also been shown that user characteristics may moderate the effect of interactive functionalities on the evaluation of explanations [71]. Regardless of its type, an explanation may not satisfy all possible explainees [103]. Moreover, individual user characteristics can influence the evaluation of a RS [61, 120], for which we assumed that this would also be the case for explanations, as discussed by References [7, 46, 50, 62].

[2] https://cloud.google.com/dialogflow.

Since a main objective of providing explanations is to support users in their decision-making, investigating the effect of differences in user characteristics with respect to the decision-making process is of special interest to us. As discussed in References [46, 50], decision-making styles are determined significantly by preferences and abilities to process available information [35]. Particularly, we focus on the moderating effect of the *rational* and *intuitive* decision making styles [41], the former characterized as a propensity to search for information and evaluate alternatives exhaustively, and the latter by a quick processing based mostly on hunches and feelings. We also considered *visualization familiarity*, i.e., the extent to which a user is familiar with graphical or tabular representations of information.

3 CONCEPT: EXPLANATIONS AS INTERACTIVE ARGUMENTATION

To evaluate our research questions, we designed an interaction scheme for the exploration of explanatory arguments in review-based RS, as initially introduced in Reference [50]. A recommendation issued by a RS can be considered a specific form of a claim, namely, that the user will find the recommended item useful or pleasing [33, 50]. The role of an explanation is thus to provide supportive evidence (or rebuttals) for this claim [50]. Claims are, however, also present in the individual user's rating and opinions, which may require explaining their grounds as well, thus creating a complex multi-level argumentative structure in an explainable RS [50], a challenge also identified by Reference [38]. To formulate an explanation scheme able to support this type of structure, and as an extension of work reported in Reference [50], we considered dialog-based explanation models [77, 113, 114], in which instead of a single issue of explanatory utterances, an explanation is considered as an interactive, multi-turn process, where a user can indicate when additional arguments are required, to increase their understanding of system claims.

In this context, Walton proposes in References [113, 114] a model that involves explanation requests (user questions) and attempts at explanation (namely, system responses referred to as assertions, with no specific structure). However, Madumal et al. [77] note that argumentation may occur within an explanation, and model the switch between an explanatory dialog and an argumentative one, specifying the explanatory loops that can be triggered as a consequence of follow-up questions. Allowed moves within an explanatory interaction can be defined on a basis of such dialog models. However, they offer little indication of how the arguments within the moves should be structured, so their acceptance by users can be increased. To this end, and as reported in Reference [50], we based on the scheme by Habernal et al. [40], an adaptation of the Toulmin model of argumentation [108], formulated to better represent the kind of arguments usually found in user-generated content. The scheme by Reference [40] includes the following components: claim (argument conclusion), premise (overall reason to accept the claim), backing (additional specific evidence supporting the claim), rebuttal (statement attacking the claim, representing an "opposing view"), and refutation (statement attacking the rebuttal).

Our proposed overall scheme is shown in Figure 1. Unlike Walton, who modeled explanatory movements as explanation requests and attempts, we considered an explanation as an iterative process of *argumentation attempts* (the system intends to provide arguments to explain something) followed by *argument requests* (the user asks the system to provide—follow-up—arguments that support the claim that user will find the recommended item useful) [50]. With this slight renaming of the explanation components, we aim to make explicit that, unlike Walton's model, statements within system's attempts at explanation follow an argument structure. Thus, in our approach, argumentation attempts include premises (overall reason to accept the claim that a recommended item is worthy to be chosen, e.g., "Good food, great staff"), backing (e.g., percentage of positive opinions about an aspect, or actual customer comments supporting the aggregate

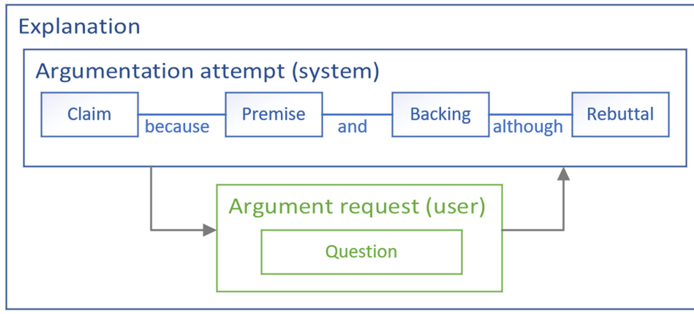

Fig. 1. Overall scheme for explanations as interactive argumentation in review-based RS. Argumentation attempts (in blue) are constituted by different argument components (claim, premise, etc.)

percentages), and rebuttal (e.g., percentage of negative opinions about an aspect or the actual negative comments by customers) [50].

We operationalized our proposal by implementing a RS using two types of interfaces or interactive paradigms: GUI navigation (see Section 5) and natural language conversation (see Section 6).

Our proposal of explanations as interactive argumentation fosters the interactive features defined by Reference [71]: active control (voluntary actions that can influence the user experience) and two-way communication (the ability of two parties to communicate with each other). These features are reflected in our proposal by the possibility to examine explanatory information at will, e.g., to access excerpts from customer reviews supporting system claims (e.g., an item is worth choosing), and to indicate to the system which aspects are of greater relevance to the user, so that content is presented accordingly. Consequently, our proposal opposes a single static view that may include aspects or details that are not relevant, or leave out those that are, while leveraging interactivity to facilitate access to structured and relevant information extracted from the reviews and to present it in explicit connection with the explanations provided by the system.

4 BASIC COMPONENTS OF EXPLAINABLE REVIEW-BASED RS

User and data collection studies reported in this article were conducted using a review-based RS, implemented to generate both recommendations and explanations, according to our proposed explanation scheme, and under different types of interface. Implementation details of this base component, which were originally reported in Reference [50], are described below.

4.1 Dataset for Aspect-based Sentiment Detection

We used the ArguAna dataset [111] for the detection of the aspect addressed in a statement and the sentiment expressed about it. ArguAna includes hotel reviews and ratings from TripAdvisor, where sentiment and explicit features are annotated sentence-wise. We further categorized the explicit features in 10 general features (room, price, staff, location, facilities, bathroom, ambience, food and beverages, comfort, and checking), with the help of two annotators (Krippendorff's alpha of 0.72), as reported in Reference [50], aiming to train a classifier to detect the main aspect addressed in a sentence (e.g., "I loved the bedding" would be classified as *room*).

4.2 Explainable RS Method

We implemented the **Explicit Factor Model (EFM)** [123], a review-based **Matrix Factorization (MF)** method to generate both recommendations and explanations, as reported in Reference [50]. The rating matrix (ratings granted by users to items) consisted of 1,284 items and 884 users extracted from the ArguAna dataset (only users with at least five written reviews were included), for a total of 5,210 ratings. Item quality and user preference matrices were calculated using the sentiment detection described previously. Each element of the item quality matrix contains the measurement of the quality of the item with respect to each aspect, while the elements of the latter measure the extent to which the user cares about an aspect. The number of explicit features was set to 10, which seemed to describe the most relevant aspects as described in Section 4.1. Model-specific hyperparameters were selected via grid-search-like optimization. After 100 iterations, we reached an RMSE of 1.27, a metric used to measure the differences between dataset values and the values predicted by the RS model. Predicted ratings were used both to sort recommendations, and are shown within explanations (average hotel rating with 1–5 green circles shown in recommendations list). The values of the quality matrix were used to calculate the percentages of positive and negative comments on aspects.

4.3 Aspect-based Sentiment Detection

We trained a BERT classifier [30] to detect the general feature addressed within a sentence: We used a 12-layer model (*BertForSequenceClassification*), 6,274 training sentences, 1,569 test sentences, and F-score 0.84 (macro average), as reported in Reference [50]. We also trained a BERT classifier to detect the sentiment polarity, using a 12-layer model (*BertForSequenceClassification*), 22,674 training sentences, 5,632 test sentences, and F-score 0.94 (macro average). The classifier was used to (1) obtain the quality of hotels (item quality matrix in EFM) and relevance of aspects to users (user preference matrix in EFM), and (2) to present participants with negative and positive excerpts from reviews regarding a chosen feature during the explanatory interaction, as elaborated in Reference [50].

4.4 Personalization Method

To overcome implications of the *cold start* problem [99] (system does not have enough information about the user to generate an adequate profile and thus, personalized recommendations), participants were asked beforehand (as part of the respective user study) for the five hotel features that mattered most to them, in order of importance. The system calculated a similarity measure to detect users in the EFM preference matrix with similar preferences. Next, the user in the matrix with the preferences most similar to those of the current user was used as a proxy to generate recommendations, i.e., we selected the proxy user's predicted ratings, and use them to sort the recommendations and features within the explanations that were shown to the current user.

5 GUI NAVIGATION

We implemented a user interface under the GUI navigation paradigm, grounded on the base RS described in Section 4. We first unfolded the explanation scheme addressed in Section 3, as shown in Figure 2, to design the explanatory interaction flow. Consequently, the realization of the scheme as a user interface is depicted in Figure 3, as originally reported in Reference [50].

As discussed in Reference [50], an explanatory dialog can take place both through verbal interactions and through a visual interface (non-verbal communication, or a combination of verbal and visual elements) [77, 82]. As for presentation, while arguments are usually associated with oral or written speech, arguments can also be communicated using visual representations (e.g., graphics or images) [9]. Consequently, as reported in Reference [50], the GUI navigation interface involves the presentation of arguments of the type "% of positive and negative opinions" using a bar-chart, implementation we used for the user evaluation together with the natural language interface (Section 6), as part of the user study reported in Section 7.

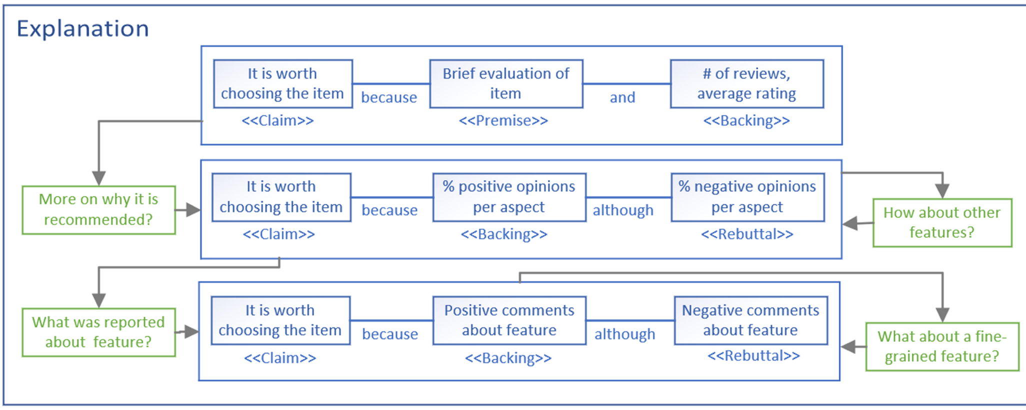

Fig. 2. Unfolded scheme for explanations as interactive argumentation in review-based RS, reported in Reference [50]. Blue boxes: argumentation attempts by the system, green boxes: argument requests by users.

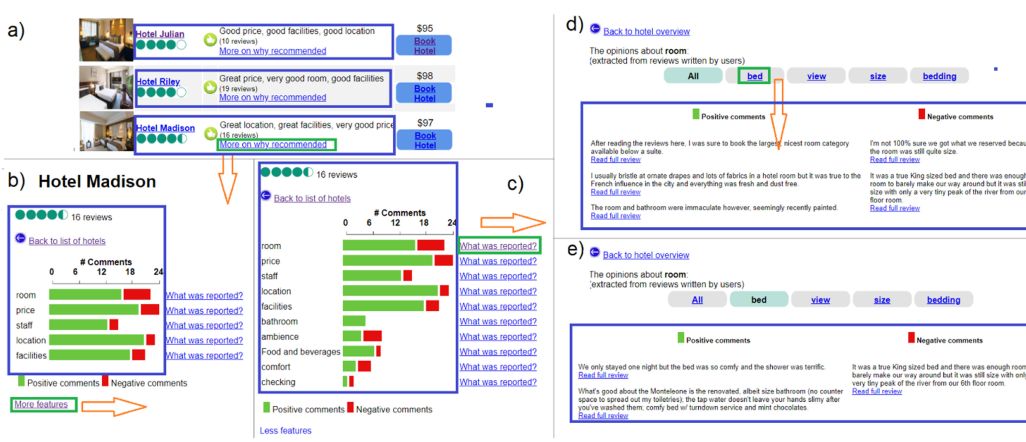

Fig. 3. Interactive explanations through a GUI navigation interface (screenshots of implemented system reported in Reference [50]). Enclosed in blue: argumentation attempts; in green: argument requests. Orange arrows: sequence of allowed moves, pointing to the next interface. (a) List of recommended items; clicking on "More on why recommended" displays: (b, c) aggregation of comments by aspect; clicking on "What was reported?" displays: (d) comments on chosen aspect; clicking a fine-grained feature button, displays: (e) comments on chosen feature.

6 CONVERSATIONAL AGENT FOR EXPLAINABLE REVIEW-BASED RS

In this section, we present our methodological approach to providing explanations as natural language conversation, and describe the steps we took for the implementation of a CA that provides such explanations. First, we report the results of a WoOz pre-study [48], which aimed to address the question of how RS users communicate their explanation needs using a CA. We then report details on the collection and annotation of ConvEx-DS [49] (a dataset for the automatic detection of question intent with explanatory purpose, as a prerequisite for the development of our CA). During corpus collection, we also measured users' evaluation of the helpfulness of the answers that the system generated automatically to their questions, results that are also reported in this section. Finally, we describe the design and implementation details of ConvEx, a CA for

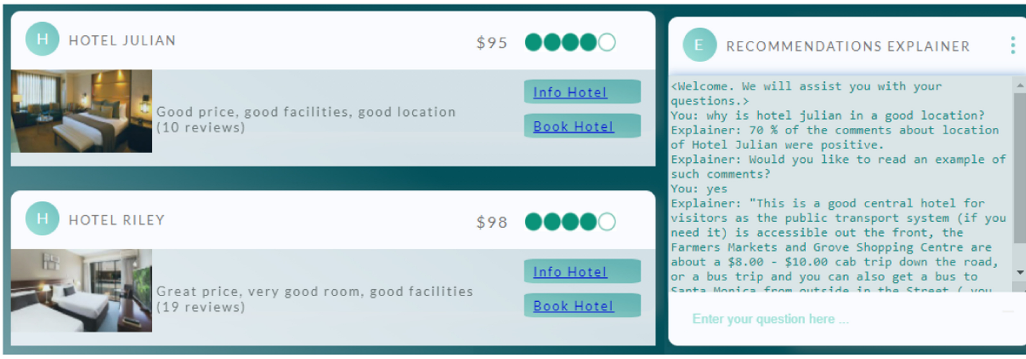

Fig. 4. User interface presented to participants in WoOz pre-study [48].

explainable review-based RS, which we used for the user evaluation reported in Section 7, under the natural language interface condition.

6.1 Wizard of Oz Pre-study

In this section, we describe the details of the pre-study that led to the formulation of our dimension-based intent model. The content of this section was initially reported in Reference [48].

6.1.1 Background. The design of adequate conversational explanations requires a proper understanding of possible user requests [48, 67]. Prior work by Reference [66] presented a dataset consisting of written conversations between humans with a movie recommendation goal, however, no explanatory inquiries like "Why do you recommend X?" are addressed. Bjerva et al. [8] collected a QA dataset for several domains (including hotels), which can be used to generate answers to factoid and subjective questions (e.g., "How is it the location?"), in regard to a specific item, but leaving out comparison and why-recommended queries. However, Liao et al. [67] formulated a XAI question bank, with questions users might typically ask about AI algorithms, but targeting users with expert knowledge in AI, whereas no similar question bank definition has been developed, to our knowledge, for end users and, in particular, for RS. Consequently, we first conducted a pre-study using the WoOz [58] method, to capture the possible questions users would ask in the context of RS explanations, particularly in the hotel domain, and a subsequent online study to collect a larger number of user-generated questions.

6.1.2 Methods. WoOz is a technique that has been widely adopted for human computer interaction prototyping [29, 78], in which a member of the research team (the *wizard*) simulates the response actions of the system, through a computer-mediated interface. We aimed, with this pre-study, to explore how users would interact with a conversational interface for explanations in RS. Here, we used as a basis the RS described in Section 4, and a dialog guideline for the wizard (see Appendix A.1, Figure 16), based on the concept of explanations as interactive argumentation (Section 3), aiming to achieve a structured conversation similar across participants. We recruited 20 participants through Amazon Mechanical Turk, and collected a total of 20 dialogues and 105 user utterances. Participants were requested to interact with a RS, and to type any question of interest about one or more hotels in the chat box located on the right of the hotel list (Figure 4); we underlined that the chat box was designed to explain the reasons for the recommendations, to prevent the user from asking questions about other processes, such as booking assistance.

In respect of data analysis, we follow an inductive category formation [79]. This step involved two independent annotators, who came to a Cohen's kappa = 0.91, almost perfect agreement

Table 1. Most Frequent Domain Intents (Combination of Dimensions Values) Sorted by Number of Questions Per Intent (Desc.), as Reported in Reference [48]

Scope	Comparison	Assessment	Detail	Example	# Qs	Type of initial response
Single	Non-comp	Factoid	Aspect	Does Hotel Julian have a pool?	29	Y/N or value
Single	Non-comp	Why-recomm	Overall	Why is Hotel Julian my top recommendation?	14	Because [Argument backing]
Single	Non-comp	Evaluation	Aspect	How is the food at Hotel Evelyn?	8	[Argument claim], because [Argument backing]
Indefinite	Comparative	Evaluation	Aspect	Which hotel has the best customer service?	7	Hotel X, because [Argument backing]
Indefinite	Non-comp	Factoid	Aspect	Do any of the hotels provide free breakfast?	6	Y/N or value
Tuple	Non-comp	Factoid	Aspect	what are the checking in times for hotel Owen and hotel Evelyn?	4	Y/N or value
Indefinite	Comparative	Evaluation	Overall	Which hotel has the best reviews?	4	Hotel X, because [Argument backing]
Indefinite	Non-comp	Evaluation	Aspect	what rooms would be good for parents with children?	3	Hotel X, because [Argument backing]
Tuple	Comparative	Evaluation	Overall	What is difference between hotel Evelyn and hotel James?	2	Hotel X has better comments on [feature x] and [feature y].

intercoder reliability [65]. To analyze follow-up questions, we validated anaphoras ("a linguistic form whose full meaning can only be recovered by reference to the context" [96]) and ellipsis ("an omission of part of the sentence, resulting in a sentence with no verbal phrase" [96]) , using criteria from References [10, 96]. See Reference [48] for further details on the WoOz setup, participants demographics and payment, and subsequent data analysis.

6.1.3 Results—Dimension-based Intent Model.

Intents and entities. We found that users' intents could be classified into two main types: *domain-related* intents (regarding hotels and their features) and *system-related* intents (regarding the algorithm, or the system input) [48]. In turn, domain-related intents could be categorized according to the following dimensions [48]:

- *Scope*: Whether the question refers to a single item (*single*), a limited list of items (*tuple*), or to no particular item (*indefinite*).
- *Comparison*: Whether the question is (*comparative*) or not (*non-comparative*). We adopt the comparative sentence definition by Reference [56] "expresses a relation based on similarities or differences of more than one object," including superlatives and relations like "greater" or "less than."
- *Assessment*: Whether the question refers to the existence or characteristics of item features (*factoid*), to a subjective assessment of the item or its features (*evaluation*), or to system reasons to recommend an item (*why-recommended*).
- *Detail*: Whether the question inquires for a specific aspect or feature (*aspect*), or for the overall item (*overall*).

Consequently, the intent of a single domain question could be defined as a combination of the four dimensions. Table 1 provides intent examples, their frequency in collected utterances, and the suggestion of possible answers that the system could provide. All but one of the questions could be classified as system-related intent, namely, "why are there so few reviews?" All questions of domain intent regarded the entities: *hotel* and *hotel feature*.

Follow-up questions. We found that 55% could be classified as standalone questions and 26% as follow-up questions. A special case are inquiries that could work as both types (19%). Such is the case for comparative questions under the value "Indefinite" of dimension *scope*, which may refer to the best among all possible options (e.g., "which is the best hotel?") or, if a subset of options was previously discussed, as a follow-up, (e.g., "I am choosing between the Riley and the Evelyn. Which is the best hotel overall?").

In regard to the follow-up question types: 24% corresponded to pronoun or possessive adjective anaphora (e.g., "I'm looking for facts about current internet service—is *it* unchanged or upgraded?"), 71% noun phrase anaphora (e.g., "When was the last time *the Hotel* underwent a remodel?), and 5% ellipsis (e.g., "what are the checking in times for hotel owen? (sic) *and hotel evelyn?* (sic)"). We noted that pronouns and noun phrases in anaphora referred only to hotels names or hotel features. Moreover, 11% of utterances contained non-question sentences aiming to establish a context for a subsequent question, e.g., "I like the ambiance of the Hotel Evelyn, how were the reviews for that?" Finally, only 2.4% of utterances contained more than one question.

6.2 Collecting a Corpus for Intent Detection: Background and Methods

In this section, we describe the details of the data collection that led to ConvEx-DS consolidation. The content of this section was initially reported in Reference [49].

6.2.1 Background.

Intent detection and slot filling. Intent detection seeks to interpret the user's information need expressed through a query, while slot filling aims to detect which entities—and also features of an entity—the query refers to Reference [49]. In the open-search domain, intent classification is the most common approach for intent representation [109]. Here, a user query can be categorized using a classification scheme: a set of dimensions or categories [12, 109]. Our proposed intent model [48] corresponds to this representation type.

As discussed in Reference [49], a large body of previous work has addressed the task of intent detection, both for open search domains (see References [52, 109]) and task-oriented dialog systems, for processes such as flight booking, music search or e-banking, e.g., Reference [17, 27, 44]. Methods proposed to solve these tasks range from conventional text classification methods, to more complex neural approaches, based on recurrent neural networks, attention-based mechanisms and transfer learning (we refer readers to the survey on this matter by Louvan and Magnini [76]).

Our work is related to the text classification approach, leveraging the representation of intents according to a classification scheme. Thus, the complex intent detection task can be split into small text classification tasks, to detect the class that best represents a sentence, for each dimension. To this aim, we leverage the language model BERT [30]. We solve the slot-filling task as a **named entity recognition (NER)** task as in Reference [39], to detect entities referred to in sentences (i.e., hotel names), for which we used the **natural language toolkit (NLTK)** [75].

Datasets for Intent Detection. Given that, to our knowledge, there were no datasets to support the intent detection task in the specific context of explainable RS, we analyzed datasets that could support text classification according to the different dimensions of the intent model:

- Dimension comparison: Jindal and Liu [56] released a dataset for classification of comparative and non-comparative sentences, using news articles, reviews, and forums on electronics and diverse brands, as a source.
- Dimension assessment: Bjerva et al. [8] released SubjQA, to detect subjectivity of questions in QA tasks, for different domains, including hotels. This dataset does not address *why-recommended* questions.
- Dimension detail: To our knowledge, no dataset addresses explicit annotation of sentences addressing overall item quality (e.g., "how good is Hotel x?"), in contrast to aspect-based sentences (e.g., "how good is the food").

Table 2. Inter-rater Reliability of ConvEx-DS, as Reported in Reference [49]

Dimension	Fleiss' kappa	Class	% of full agreement
Comparison	0.72	Comparative	77.28%
		Non-comparative	86.86%
Assessment	0.65	Factoid	73.99%
		Evaluation	58.56%
		Why-recommended	66.42%
Detail	0.75	Aspect	95.73%
		Overall	66.82%

Fleiss' kappa refers to each dimension, and the *% of full agreement* to each class, i.e., percentage of questions in which all annotators agreed on assigning that class.

- Dimension scope: Values of this dimension can be inferred from entity detection (e.g., number of entities referred), for which NER can be used.

6.2.2 Methods.

System for data collection. The RS described in Section 4 was extended by adding a NLU module to detect users' intent, using text classifiers trained initially on the auxiliary datasets described above ([8, 56], and data collected in Reference [48] for dimension detail), as well as a module to generate answers consistent with the method on which the RS is based (EFM [123]). Development and corpus collection involved a process consisting of several iterations, as reported in detail in Reference [48], aiming to refine methods for answer generation. For example, early iterations included methods to respond to the most common user intents; new iterations included responses to new occurrences of intents observed in the previous iteration.

Intent detection. As reported in Reference [49], we trained BERT classifiers [30], one for each dimension (comparison, assessment, and detail), using a 12-layer model (*BertForSequenceClassification*, bert-base-uncased), batch size 32, and the Adam optimizer (learning rate = $2e-5$, epsilon = $1e-8$). Classifiers for comparison and assessment converged after four epochs, while for detail five epochs were needed. Datasets were split randomly into training (80%) and test (20%) during the training phase. To avoid overfitting, the most represented class was downsampled (randomly) to approximate the size of the less represented class, which was slightly upsampled (randomly) to fit round numbers like 1,000 and 500. In the case of the detail dimension, due to the small size of the auxiliary dataset, both classes were increased to 100 instances each (using procedure described in Reference [49]). Datasets (original and balanced) sizes are reported in Table 2.

Answer generation module. We implemented a module to generate replies based on the intent detected. According to this proposal, *factoid* questions could be replied with Y/N or a value (e.g., check in times) based on metadata. As for *evaluation* or *why-recommended* questions, replies were based on the aggregation of positive or negative opinions regarding an aspect (if the question was *aspect*-based), or the most important aspects for the participant (in case the question was of type *overall*). If the question was comparative, then the system calculated which hotel was better among a tuple, or the best in general (scope *indefinite*), based on the aggregation of the opinions. These are some examples of the type of responses generated by the system: Q: "Why does Hotel Hannah have the highest rating?" A: "Because of the positive comments reported regarding the aspects that matter most to you: 86% about location, and 85% about price."; Q: "Which hotel is best, Hotel Lily, Hotel Amelia or Hotel Hannah?" A: "Hotel Lily has better comments on the aspects that are most important to you (location, facilities, staff). However, Hotel Amelia has better comments about room, price."; Q: "Hotel Amelia is described as having a great room, what makes it great?" A: "Comments about rooms are mostly positive (90%)."

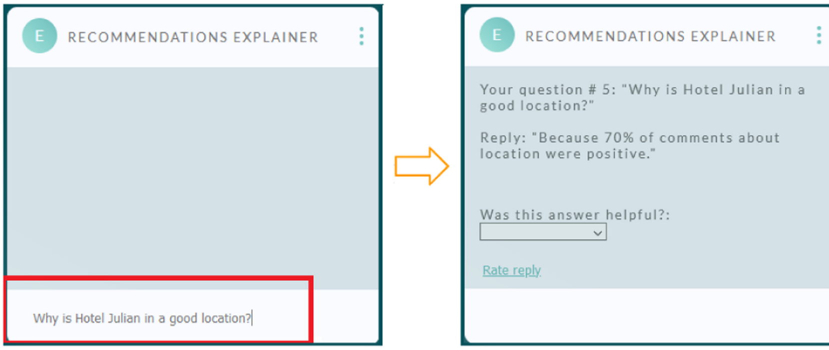

Fig. 5. User interface for corpus collection, as reported in Reference [49]. Box to write questions in red. Then, system shows the answer and requests user to rate helpfulness. (Recommendations list was displayed on the left side as in Figure 4).

6.2.3 Corpus Collection Study. We recruited 298 participants through the crowdsourcing platform Prolific, who interacted with the RS, and typed their questions about the reasons why the hotels were recommended, using their own words. We stated in the instructions that "The aim of the system is to provide explanations based on your questions" (to prevent the user from asking questions about other processes, such as booking assistance). For each question, and once the system response was displayed, users were asked to rate the helpfulness of the answer. Further details on study setup, participants demographics, and responses quality check are reported in Reference [49]. A total of 1,836 users' written questions were collected in this step, see Figure 5.

6.3 Corpus Annotation

In this section, we describe the details of the data annotation that led to definition of the gold standard for ConvEx-DS. The content of this section was initially reported in Reference [49].

6.3.1 Intent Type Annotation. Sentences where first classified according to the types: domain-related intents (regarding hotels and their features), and system-related intents (regarding the algorithm, the input, or system functionalities). Given that we detected that domain questions clearly outnumbered system questions in our WoOz study, research team members annotated this class (98.3% agreement), as the low number of system-related questions could lead to the category being ignored in the crowdsourcing setup. Disagreements were resolved in joint meetings.

6.3.2 Dimension-based Annotation. Domain-related sentences were used for the dimension-based annotation. We collected annotations for comparison, assessment and detail as independent tasks. The dimension scope was not annotated under the proposed procedure (is not a classification task but a NER task).

Annotators and crowdsourcing setup. Every sentence was annotated by three annotators: one member of the research team and the other two crowdsourced on the Prolific platform. We divided the set of questions into 19 blocks of 100 sentences each, and every block had to be annotated for each dimension separately. The annotator from research team annotated all blocks for the three dimensions, while different crowdsourcing workers could annotate different blocks for different dimensions (same annotator did not annotate the same block for the same dimension more than once). Each block included five attention checks, and annotation guidelines remained visible during the task (see Appendix A.2). A total of 92 Prolific workers performed the task, but only responses of 57 workers were used for the calculation of inter-rater reliability and the deduction of gold standard, after applying the quality check. Further details on study setup, participants demographics, payment, responses quality check, and processing are reported in Reference [49].

6.4 Corpus Collection and Annotation: Results

We report in this section the annotation statistics and results of the validation of the intent model, which included evaluation of helpfulness of answers to user questions and evaluation of the performance of classifiers trained and tested on ConvEx-DS. The content of this section was initially reported in Reference [49].

6.4.1 Annotation Statistics. Of the 1,836 collected questions, 30 were discarded (nonsense statements or highly ungrammatical), for a final set of 1,806 of annotated questions. Of these, only 24 were annotated as system-related questions. Length of questions: characters M = 39.2, SD = 15.67, words M = 7.35, SD = 2.86. We found a Fleiss' kappa of 0.72 for *comparison*, of 0.65 for *assessment* and of 0.75 for *detail*, indicating a "substantial agreement" [65], for all three dimensions. As for classes with lower percentages of questions with full agreement, we identified the following main causes:

Dimension assessment. —*Why-recommended* questions rated as subjective, given that adjectives like "good" or "great" are included in sentences, e.g., "why is hotel hannah location great?" —Questions that should be replied with a fact, but include adjectives that indicate subjectivity, e.g., "does hotel emily have any bad reviews?," "are there good transport links?," "which hotel best fits my needs?" —Questions with adjectives like "cheap," "expensive," "close," "near," "far," which can be answered with either subjective or factoid responses, e.g., "Which is the cheapest hotel?," "is there an airport near any of these hotels?" —Questions of the type "what is ... like," e.g., "What is the room quality like at Hotel Emily?" (this type of questions were actually not addressed in instructions).

Dimension detail. —Concepts that were regarded as hotel aspects, e.g., value (Which is best value for money), ratings ("What is the highest rating for Hotel Levi?"), reviews ("Which hotel has the most reviews?"), stars ("5 stars hotels?").

Finally, we have detected some questions that could hardly fit in the planned classes, e.g., "How do you define expensive? Do you compare against facilities and what is included in the price?," "The Evelyn has 17 reviews and a positive feedback but scores lower than others with less reviews. Why is this?" However, we found this number to be rather low (16 questions).

6.4.2 Helpfulness of System Answers. In the most refined version, the system was able to generate an answer in 80.58% of the cases, and to partially recognize the intent or entities in 7.34% of the cases (thus asking the user to rephrase or indicate further information). Among the main reasons why questions could not be answered we found: complexity of the question or no information available to reply (31%), text that could be improved when replying to factoid questions (23%), wrong intent classification (11%), and system errors (11%).

Figure 6 (middle) shows the evaluation of answers' helpfulness, according to ratings granted by study participants, measured on a 1–5 Likert-scale (1: Strongly disagree, 5: Strongly agree), across all iterations (M = 3.58, SD = 1.34). When taking into account only the last two iterations (which account for 63.85% of sentences collected, and involve the most refined versions of the system), we observed a greater helpfulness (to M = 3.70, SD = 1.30), which seems a reasonable value considering that the main challenge is the intent detection. We considered as "non-helpful" responses those that were marked with the values "Strongly Disagree" and "Disagree" when participants were asked "Was the answer helpful?" We analyzed the responses given to those questions by the system and

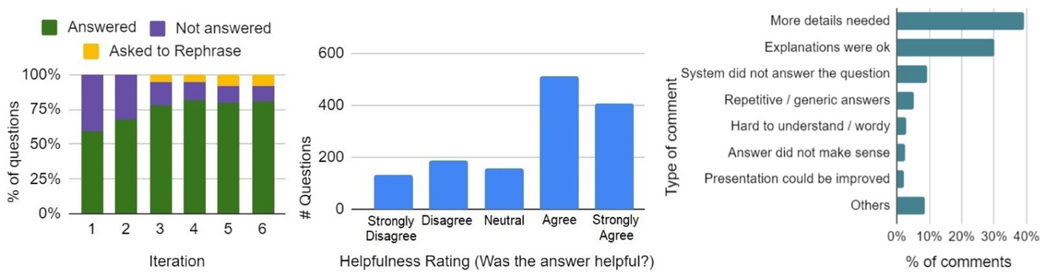

Fig. 6. Left: Distribution of replied questions, across iterations, as reported in Reference [49]. Middle: Histogram of helpfulness rating granted by users to system answers (all iterations). Right: Types of comments by participants during corpus collection, in regard to system answers.

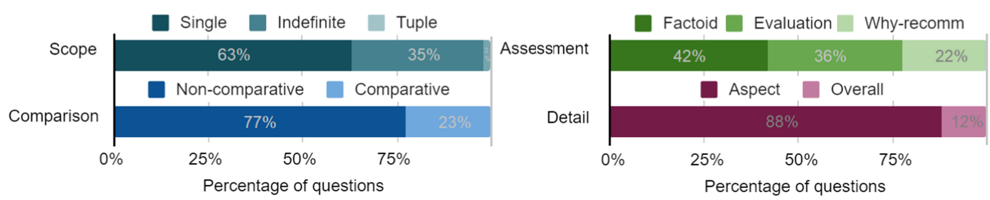

Fig. 7. Distribution of questions in ConvEx-DS (domain-related intents), as reported in Reference [49].

found that in 34% of cases, replies provided actually make sense, i.e., seemed a reasonable answer to the asked question. Among the reasons for non-helpfulness, we found: 30% due to misclassified intents or entities, 14% to system errors, 9% text that could be improved when replying factoid questions, 5% due to complexity of the question or not information to reply it, and 5% to specific aspects not addressed by the solution.

6.4.3 Intent Detection Performance.

Classifiers trained on ConvEx-DS. Details on the BERT model used, including batch size, Adam optimizer parameters, and splitting are reported in Section 6.2.2. To avoid overfitting, the most represented class was downsampled (randomly), to approximate the size of the less represented class. Comparison and assessment classifiers converged after four epochs, the detail classifier after five epochs, see Figure 7.

Dimensions comparison, assessment and detail. To verify the performance of classifiers, we calculated the F1 score, a measure of classification accuracy. We tested accuracy in three different steps: (1) performance of models trained on auxiliary datasets [8, 56], used for the system used in corpus collection. (2) We tested these models using our newly obtained annotated data, ConvEx-DS. (3) We trained and tested new classifiers, based entirely on ConvEx-DS. We report F1 scores for each dimension (*comparison*, *assessment* and *detail*). We report weighted average, to take into account the contribution of each class, which in (2) is particularly unbalanced (no downsampling of the test set was done, since balanced data was pertinent only for training).

We found that the classifier trained on Bjerva et al. [8] performed particularly poor when tested with our annotated data ConvEx-DS (see Section 6.5.3 for a discussion on the matter), see Table 3. Here, we detected that 32% of the questions marked as "evaluation" in ConvEx-DS, but classified as "non-subjective" correspond to questions regarding indefinite or more than two hotels (e.g., "which hotel has the best facilities?"), 18% corresponded to adjectives like "close," "far," "expensive," and 14% to questions of the form "what is the food like?" As of factoid questions in ConvEx-DS

Table 3. F1-scores (Weighted Average) of Classifiers of Different Dimensions, Trained and Tested on both Auxiliary Datasets and ConvEx-DS, as Reported in Reference [49]

Dimension	Dataset	F1
Comparison	Jindal and Liu [56] [Training, Testing]	0.87
	Jindal and Liu [56] [Training], ConvEx-DS [Testing]	0.88
	ConvEx-DS [Training, Testing]	0.92
Assessment	Bjerva et al. [8] [Training, Testing]	0.93
	Bjerva et al. [8] [Training], ConvEx-DS (without why-recomm) [Testing]	0.60
	ConvEx-DS [Training, Testing]	0.91
Detail	WoOz augmented [Training, Testing]	0.98
	WoOz augmented [Training], ConvEx-DS [Testing]	0.90
	ConvEx-DS [Training, Testing]	0.92

classified as subjective, we found 33% of questions involving indefinite or more than two hotels, and 32% regarded questions of the form "does the hotel have...".

Dimension scope. Entities (hotels) addressed in sentences were detected using the NLTK library. To check the accuracy of the method, two members of the research team checked the inferred entity for the collected corpus, and found that in 5.38% of cases, the inferences were wrong. Most of these cases corresponded to cities, or facilities recognized as entities, a drawback detected in early stages of corpus collection, thus additional validations were added to the procedure, so that these cases would not occur in future iterations.

6.5 Discussion: How Users Communicate Their Explanation Needs Using a CA—Validation of Intent Model

In this section, we discuss the main findings of the WoOz pre-study, and the ConvEx-DS collection and annotation process. The content of this section was initially reported in References [48, 49].

6.5.1 Types of User-generated Questions. We observed that user-generated questions collected in studies reported in References [48, 49] adhered to the explanatory objective as expected, i.e., no questions regarding other processes were asked, such as hotel booking. We also observed that users actively expressed their needs for explanation, taking the lead in formulating their own questions (not expecting the system to choose what to explain) and challenging the system's attempts at argumentation when the answers provided did not satisfy their need. As expected, participants asked factoid, evaluative, and why-recommended questions. Interests expressed in factoid questions could be handled as conditions desired by users, and as a response, the system can adapt recommendations' appearance (e.g., highlighting those that match the desired conditions). Additionally, as expected, users not only generated standalone questions but also follow-up questions, which confirms our expectation that an interactive QA approach would be appropriate to keep track of context and previously referred entities.

As discussed in Reference [49], comparing our collected inquiries with the prototypical questions from the XAI question bank by Reference [67], we found that their why-questions had a similar objective to our why-recommended: to ask for reasons why certain predictions have been provided. However, we also observed the following differences in regard to other types of questions: (1) Questions about the type of input (e.g., "what kind of data does the system learn from?") were asked only once. (2) No questions were asked about meaning of the output (e.g., "what does the system output mean"), neither on performance (e.g., "how accurate or reliable are predictions?"). We noted that how-questions asked mostly "how the opinions are" rather than asking about the overall logic of the system. (3) No "What if..." questions were asked. However, factoid questions might implicitly ask such questions (e.g., "Which hotel has a gym?" could be considered as "what if the system takes into account that "gym" is an important feature to me?"). These differences could be explained by the context of the task to be performed, and the type of users involved (general public versus AI experts). However, it was somehow surprising to us that only 24 of 1,806 of the collected questions referred to the system itself, its algorithm, or the type of input used for predictions. We believe that this may have been due to: (1) Users might have perceived that the recommendations matched their preferences and that they had generally positive opinions, i.e., they did not receive very strange recommendations that raised their suspicions. (2) Decisions in the chosen domain (hotels) are not as sensitive as in the medical or credit lending domains, where understanding the system logic or input influencing the prediction is critical to the acceptance of the system arguments. (3) The perspective and opinion of others might be more relevant than details about their own inferred profile, as reported by Reference [47] for opinionated explanations in a hotel RS.

6.5.2 Intent Model Validation. As addressed in Reference [49], we set out to validate how helpful the answers were to users, as an indirect measure of model validity. In this respect, we found that system answers were evaluated as predominantly helpful. We note, however, that ratings of non-helpfulness did not necessarily imply a mismatch of the detected intention. In fact, we found that almost one-third of responses rated as non-helpful fitted the question (i.e., made sense). After a review of participants' feedback on the system answers, we found that, although many users found them helpful or "ok," the main criticism was that some of the answers lacked sufficient detail, which is consistent with findings reported in Reference [50], that perception of explanation sufficiency was greater when options to obtain further details on system claims were offered. Thus, although the intent model seems appropriate to generate an initial or first level response, a dialog system implementation must go beyond this initial response, offering options to drill down into the details, which is consistent with our proposed scheme of explanation (Section 3), and the dialog policy introduced in Section 6.6.

In regard to the annotation task, we found a substantial agreement in all the annotated dimensions, as well as a very encouraging accuracy, when classifiers were trained on the ConvEx-DS, which leads us to conclude, that under our proposed intent model and annotation guidelines, the questions could be, to a substantial extent, unequivocally classified [49]. We note, however, the challenge of addressing the dimension assessment. In this regard, we found that the main difficulty was to classify correctly questions that could be regarded as evaluation, given their subjective nature (including expressions like "how close/far"), but for which a factual-based response could be given (e.g., "100 meters from downtown"), a similar concern raised by Bjerva et al. [8]: "a subjective question may or may not be associated with a subjective answer."

6.5.3 Performance of Intent Classifiers. We found that intent classifiers perform better when trained on ConvEx-DS, compared to classifiers trained on the auxiliary datasets, but tested on ConvEx-DS [49]. Here, the most striking case concerns the dataset for the detection of **subjective questions (SubjQA)** by Bjerva et al. [8]. The above in no way suggests anything problematic in the SubjQA itself, only that in comparison to ConvEx-DS (dimension "evaluation"), the two datasets measure rather different concepts. SubjQA addresses the subjectivity of the question asked, not whether the question involves an *evaluation* that might be subjective, as in ConvEx-DS. Thus, for example, "how is the food?" is classified as non-subjective under SubjQA, since it does not contain expressions indicating subjectivity. Thus, non-subjective under SubjQA does not necessarily imply factoid. In addition, classifiers trained in SubjQA do not work well with questions that involve some sort of comparison between multiple items, since the SubjQA only involves questions addressing single items, for which an answer could be found in a single review.

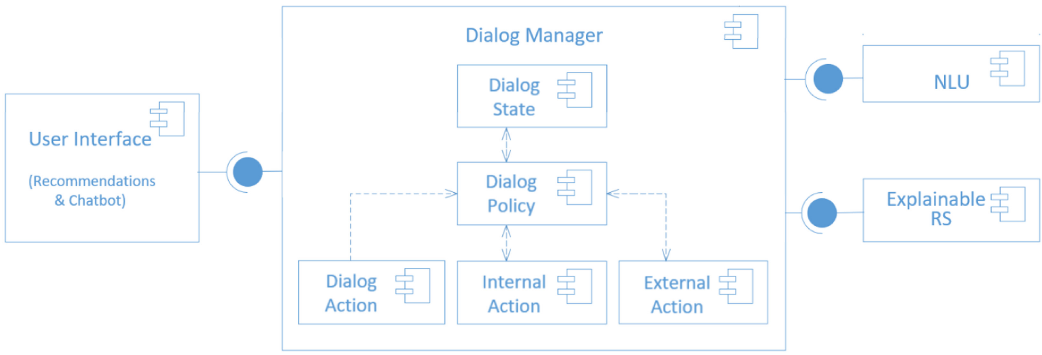

Fig. 8. ConvEx component diagram, reflecting the prevailing dialogue management architecture [42].

6.6 Conversational Agent: Implementation

We implemented ConvEx, a dialog system for explainable RS, which follows the finite-state approach, where the dialog state has a fixed set of possible transitions to other states. The architecture of ConvEx corresponds to the prevailing dialog management architecture, according to Reference [42]. Figure 8 illustrates the components involved in the solution, namely:

Dialog Manager. This component is responsible for the dialog state and flow, preserving the dialog context, and deciding on the next action to be executed by the chatbot. In turn, it involves the following components:

- Dialog State: This component allows to keep dialog context, i.e., intent and entities (hotels) addressed at each step of the dialog, for example to be able to answer follow-up questions, in case the hotel asked is not explicitly mentioned in current users' question.
- Dialog Policy: The dialog management policy defines the set of valid actions that can be executed by the chatbot given a dialog state. It is implemented in the system as an xml file, and it is depicted in Figure 9. The dialog policy can lead to the execution of different types of actions: dialog actions (to show the user a text output, e.g., the answer to a question, or to provide buttons to enable further user actions), and external actions (e.g., to invoke external REST methods, e.g., to get the list of customer comments given a hotel and a feature).

User interface. The user interface was implemented using the Python-based web framework Flask. It consists of two sections, as depicted in Figure 10: list of recommendations and information about each hotel is displayed (left) and the chatbot section (right).

Rest-APIs.

- NLU: Natural language understanding module, in charge of detecting intent, and entities. This module was trained on ConvEx-DS (see Section 6.4.3), for which we leveraged the language model BERT [30], and procedures from the library NLTK [75] to identify the entities (particularly the tokenizer and the **part of the sentence (POS)** tag methods).
- ExplainableRS: We extended the implementation of the RS detailed in Section 4, and the method described in Section 6.2.2, by exposing the following as REST methods: GetReply (to obtain answers to users questions), GetComments (to obtain excerpts from customers given a hotel, a feature, and a polarity), GetHotelsFeature (to obtain hotels that offer a certain feature), GetReviews (to obtain reviews given a hotel), and GetRecommendations (to get recommendations, given a set of preferences chosen by the user).

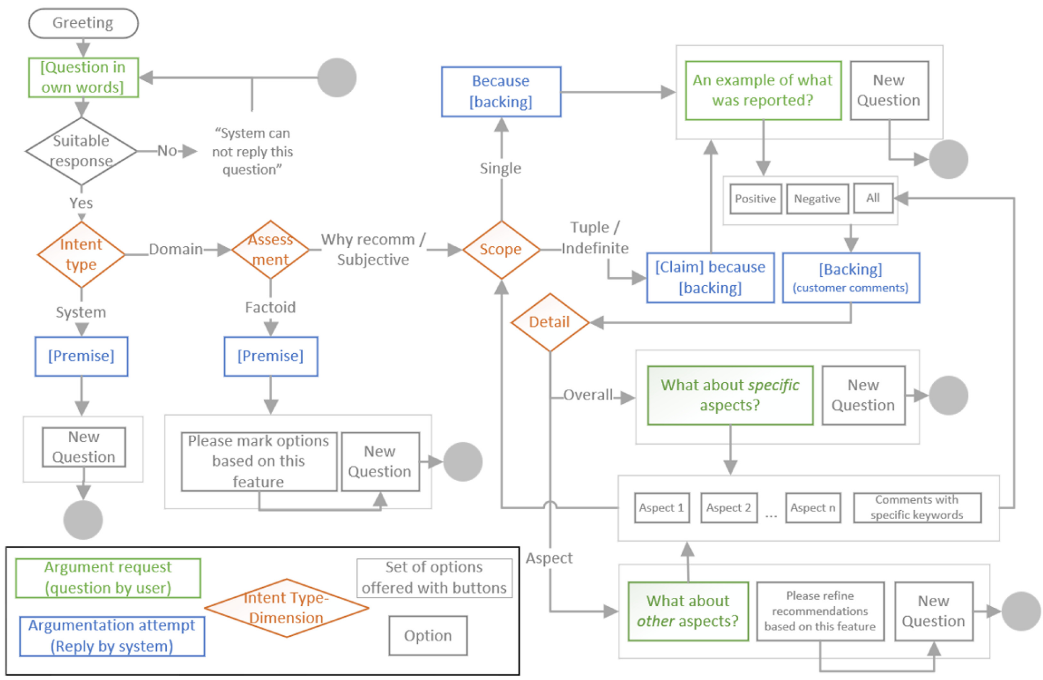

Fig. 9. Dialog management policy for conversational explanations in review-based RS.

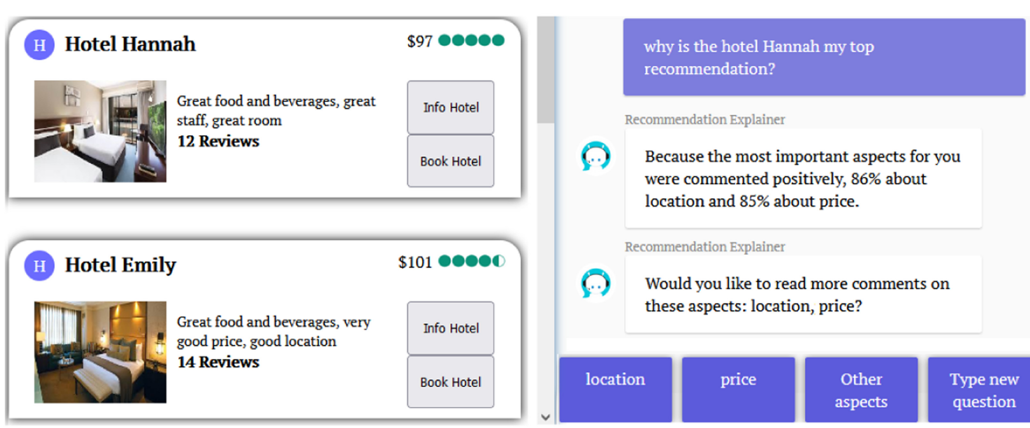

Fig. 10. ConvEx user interface.

7 EMPIRICAL EVALUATION OF THE EFFECTS OF INTERFACE TYPE AND DEGREE OF INTERACTIVITY

We conducted a user study to investigate the effect that interactivity degree and interface type may have on the evaluation of the system and its explanations, in terms of perceived explanation quality, system transparency, effectiveness, and trust, and to what extent the user characteristics (rational and the intuitive decision making styles and visualization familiarity) could influence such evaluation. To investigate how the evaluated constructs relate to each other, we evaluated a structural model (see Figure 13), as described in detail in Section 7.2.

In our analysis, interactivity degree and interface type were defined as exogenous dummy variables, while perceived explanation quality, transparency, effectiveness, and trust, as well as the user characteristics, are regarded as the latent variables.

The factors interactivity degree and interface type could be regarded as objective system aspects, according to the evaluation framework defined by Reference [61]. Possible values of *interface type* are: interactive *GUI navigation* and *natural language* conversation. Possible values of *interactivity degree* are *high* and *low*.

We emphasize that both conditions (interactivity degree high and low) can access the same amount of information, that is, both conditions have access to all the reviews reported by customers for each item, as well as to explanations showing percentages of pros and cons per aspect. What varies is the possibility of accessing structured information extracted from reviews, by means of interactive options, so that the high interactivity option allows access to excerpts from the reviews, filtered by polarity and aspect, that support the aggregated view of pros and cons. Under the low interactivity conditions, users do not have the chance to navigate from aggregated accounts (% of pros and cons) to filtered specific comments extracted from reviews.

The study follows a 2×2 between-subjects design, and each participant was assigned randomly to one of four conditions (combination of *degree* and *type*, see Figure 11). Table 4 summarizes the differences between the conditions.

According to our proposal, explanations with a higher degree of interactivity allow users to contest the arguments of the system in the form of aggregated opinions, allowing to review in detail customer comments that compose such aggregation, which could facilitate the understanding of the recommendations. Thus, we hypothesized that:

H1: A higher degree of interactivity has a positive effect on the perception of explanation quality by users.

Pu et al. [95] demonstrated causal relationships among RS evaluation constructs. They found that a positive perception of explanation quality by the user can lead to a positive assessment of transparency, which in turn can lead to a positive assessment of trust. In addition, they found that a positive perception of information sufficiency (a factor considered in our evaluation of explanation quality) can lead to a positive perception of perceived usefulness, which is related to the definition of system effectiveness by Knijnenburg et al. [61]. Consequently, we expect:

H2: Greater scores of perceived explanation quality will lead to greater scores of perceived transparency, trust, and effectiveness.

While work reported by Pu et al. [95] addresses transparency on the basis of a single item "I understood why the items were recommended to me," we use the questionnaire items to measure perceived transparency as reported in Hellmann et al. [43], and we investigate how the sub-constructs involving such items relate to each other and how they relate to constructs related to trust (trusting beliefs and trusting intentions, as defined by Reference [80]). Particularly, we address the perceived transparency constructs: information provision ("system provided information to understand how/why is an item recommended"), understanding input ("what data does the system use"), understanding output ("why and how well does an item fit one's preferences"), and interaction ("what needs to be changed for a different prediction"). As discussed by Reference [43], items under the construct information provision are linked to the availability of functions of the system, that explicitly disclose information about how recommendations were inferred. Particularly, we hypothesized:

H3: Higher scores on users' perception of the presence of these functions lead to a better understanding of the recommender's input and output.

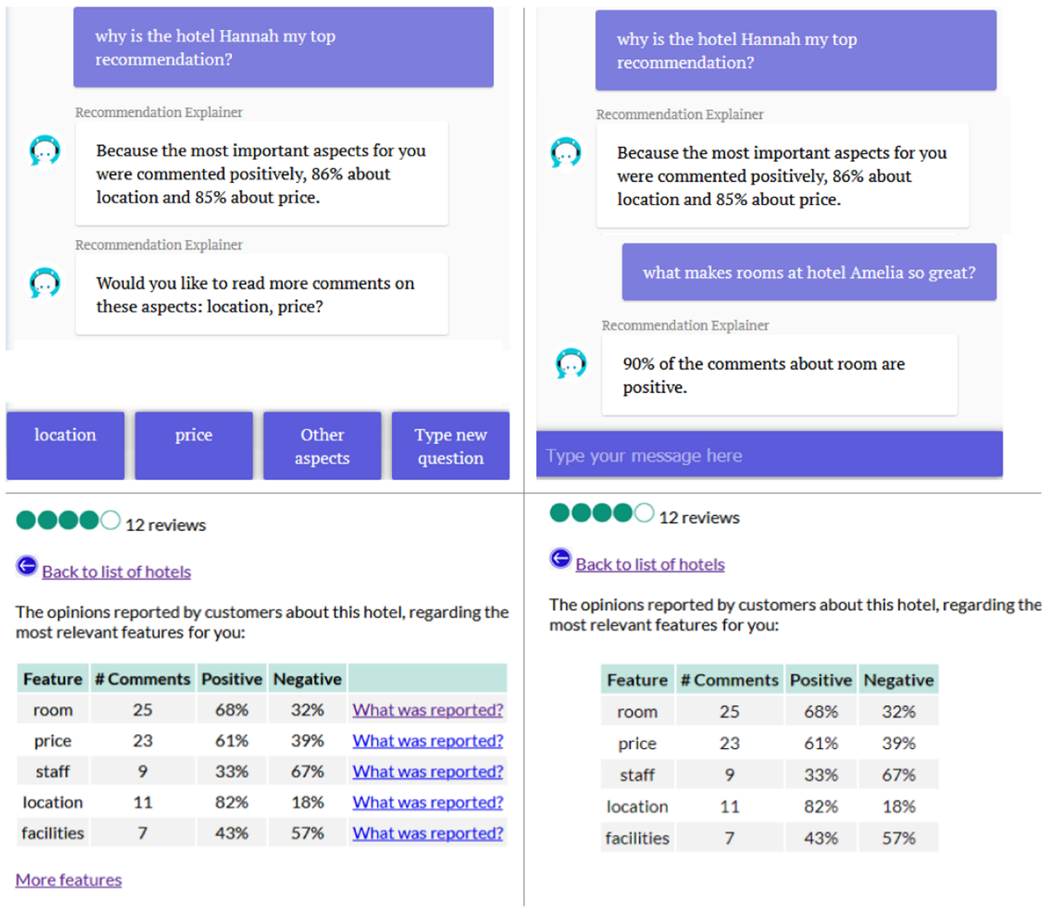

Fig. 11. The four different interfaces used in the user study. Manipulation of interface type in combination with interactivity degree. Top: natural language; bottom: GUI navigation. Left: interactivity degree high; right: low.

H4: Higher scores for perceived understanding of the recommender's input and output lead to a better understanding of what needs to be changed to get a different prediction.

Proponents of dialog models of explanation argue that providing explanations through a conversation between the user and the system, in which the user can contest the system's arguments, can facilitate user understanding of system decisions [51, 114]. This understanding, in turn, is the basis on which the concept of transparency has traditionally been evaluated in RS (e.g., References [28, 34, 95]).

Since the natural language interface allows the user to formulate a wider range of questions, even in their own words, compared to the GUI navigation approach, where questions are limited to a much smaller number of fixed options represented in links and buttons, we hypothesized that:

H5: The use of the natural language interface has a greater positive effect on the perception of transparency by users, compared to the GUI navigation.

Additionally, we hypothesized (in line with References [61, 71]) that a number of user characteristics may moderate the effect of interactive functionalities on the evaluation of the system

Table 4. Main Differences between Conditions Compared in User Study

Interface Type	
GUI Navigation	**Natural language conversation**
Questions: —Fixed (“why recommended”, “what was reported on feature”), as hyperlinks or buttons *Answers:* —Table with 5 to 10 most relevant aspects for the user, # of comments and % of positive and negative comments, per aspect. —Customer comments, filtered by aspect/features and polarity.	*Questions:* User written questions, using own words. *Answers:* —Depending on detected intent (details in Section 6.2.2). Answers to subjective and why-recommended questions involve % of positive and negative comments per aspect. —Customer comments, filtered by aspect/features and polarity. —Refined recommendations, for comparative and factoid questions.
Interactivity degree	
High	**Low**
—User can access arguments (customer comments), which support explanation attempts with % of positive/negative opinions per aspect. —Refined recommendations (filter hotels offering a feature), when questions are factoid (only ConvEx)	—User can not access to filtered customer comments associated to explanations. —nor filtering hotels offering a feature.

and its explanations. Particularly, users with greater visualization familiarity have a greater ability and habit of processing aggregated information through graphical formats, such as those included in the GUI navigation condition, and users with lower visual abilities might benefit less from a presentation based on tables or graphics [59, 100]. Additionally, given the propensity of the most intuitive users for rapid information processing, based primarily on hunches and feelings [41], we hypothesized a preference of such users for explanations through graphical formats, since these facilitate a quick processing of information, given to their greater immediacy (possibility of quick processing) [9]. Thus, we hypothesized that:

H6: More intuitive users, and users more familiar with data visualization, will evaluate GUI navigation explanations more positively.

Since users with a predominant rational decision-making style have a preference for evaluating information extensively during decision making [41], as reported by Reference [50], and facilitated by our argumentative explanation design, we hypothesized that:

H7: In the high interactivity condition, users with a predominantly rational decision style will evaluate explanation quality more positively than less rational decision makers.

7.1 User Study

7.1.1 Questionnaires.

Evaluation. We used the following items to address perception of explanations, which we will refer to as *perceived explanation quality* (internal reliability Cronbach’s: $\alpha = 0.87$): items from the **user experience items (UXP)** of Reference [62], i.e., confidence (explanations makes user confident that she/he would like the recommended item), transparency (explanation makes the recommendation process clear), satisfaction (user would enjoy a system if explanations are presented this way, item adapted from the original), and persuasiveness (explanations are convincing); an item adapted from Reference [32] (explanations provided are sufficient to make a decision) to evaluate explanation sufficiency; and an item from Reference [47] (explanations are easy to understand).

To measure users’ evaluation of the RS transparency, we used items proposed by Reference [43] involving the sub-constructs: information provision (“system provided information to understand how/why is an item recommended”), understanding—input (“what data does the system use”), understanding—output (“why and how well does an item fit one’s preferences”), and interaction (“what needs to be changed for a different prediction”).

We utilized items from Reference [61] to evaluate perceived effectiveness, construct *perceived system effectiveness* (system is useful and helps the user to make better choices). We adapted items from Reference [80] to evaluate trust in the system: constructs *trusting beliefs* (user considers the system to be honest and trusts its recommendations), and *trusting intentions* (user would confidently choose an option based on recommendation, and would be willing to share information).

All items were measured on a 1–5 Likert-scale (1: Strongly disagree, 5: Strongly agree). In addition, the following open-ended questions were asked: "Please provide the reasons why you chose the hotel you did. A bonus of up to £0.3 will be paid depending on the quality of this response," "Please let us know about your overall opinion about the explanations provided or how they could be improved," and "How would you explain to a friend, in your own words, how the system generates recommendations?" Questionnaire items used in user study are listed in Appendix B.

User characteristics. We used all the items of the decision style scale proposed by Reference [41], which is a well-validated instrument for rational and intuitive decision making. We used the visualization familiarity items proposed by Reference [62] ($\alpha = 0.87$). All items were measured with a 1–5 Likert-scale (1: Strongly disagree, 5: Strongly agree). Questionnaire items are listed in Appendix B.

7.1.2 Participants. We recruited 223 participants (110 female, mean age 35.4 and range between 18 and 80) through the crowdsourcing platform Prolific. We restricted the task to workers in the U.S. and the UK, with an approval rate greater than 98%. Participants were rewarded with £1.7 plus a bonus up to £0.30 depending on the quality of their response to the question about reasons why they chose a certain hotel, set at the end of the survey, aiming to achieve a more motivated hotel choice by participants, and encouraging effective interaction with the system. Time devoted to the task (in minutes) was M = 17.05, SD = 6.22 (time of interaction with the system was M = 6.22, SD = 3.78).

We applied a quality check to select participants with quality survey responses. We included attention checks in the survey, e.g., "This is an attention check. Please click here the option 'Disagree'." The responses of 37 of the 270 initial participants were discarded, on the basis of failing validation checks, suspicious response patterns (e.g., the same score for all questions on the same page) or a very low time using RS (less than 40 s). Additionally, 10 rows were detected as outliers, based on Cook's distance, which were subsequently dropped, for a final sample of 223 subjects.

7.1.3 Procedure. First, the participants answered the questionnaires on user characteristics. They were asked to report their five most important aspects when looking for a hotel, sorted by importance. They were presented with instructions on how to complete the task: (1) A list of 10 recommended hotels would be presented (as a result of a hypothetical hotel search already performed using a RS, i.e., no search filters were offered). (2) They had to evaluate the different options presented, come to a decision on which hotel appeals to them the most, and back in the survey, they had to indicate the reasons for the chosen option.

They were also presented with specific instructions and screenshots on how to interact with the system and obtain explanatory information, depending on the condition assigned (these were also included within the system as a reminder). In the conditions with a natural language interface, a list of nine examples of the type of questions they could ask to the chatbot was provided but participants were also informed that they were completely free to formulate their own questions. Providing examples was done to avoid obvious out-of-scope questions, such as price- or booking-related questions. The examples were similar to those provided to annotators during ConvEx-DS corpus annotation (see Appendix A.2). Participants were then presented with a cover story, to establish a common starting point in terms of travel motivation (a holiday trip). The participants then used the application and filled out the system evaluation questionnaire.

7.2 Results and Structural Equation Model (SEM)

To assess the effects of the experimental conditions and to obtain a better understanding of the relationships between the different factors involved, we set up a Structural Equation Model (SEM). We estimated the lower-bound for the sample size, based on the number of the latent constructs and the observed variables. The minimum sample size required is 184 to at least detect medium effects (0.3), with probability level set to $\alpha = 0.05$ and a desired statistical power level of 0.8 [25, 118].

We conducted outlier detection, a normality test, and the selection of an appropriate estimator, prior to the evaluation of the model. Shapiro's test for normality pointed to variables of interest that significantly deviated from normal distributions, so we used a MLM estimator for our analysis, which allows for robust standard errors in case of non-normality and scaled test statistics [37]. Interaction effects (between user characteristics and explanation quality, and between experimental conditions and explanation quality) were tested by including interaction terms in the SEM model.

Descriptive results of our study are included in Table 7, split by interface type and interactivity degree.

Prior to performing the SEM, we validated the factor structure of the set of observed variables, by testing internal reliability and convergent validity, using **Confirmatory Factor Analysis (CFA)** (see Table 5). We also evaluated the discriminant validity of the resultant set of items (see Table 6).

7.2.1 Convergent and Divergent Validity. Cronbach's alpha values for all constructs exceeded the 0.5 limit considered as acceptable [93]. Items with loadings lower than 0.5 (acceptable level, according to Reference [93]) were removed. CFA model fit of (CFI = 0.915; TLI = 0.906; RMSEA = 0.047; SRMR = 0.057). Finally, the average variance extracted (AVE) of all but three constructs is higher than 0.5, a value commonly accepted as an indicator of convergent validity (see Reference [36]). However, constructs with AVE values above 0.4 whose composite reliability is above 0.6 can still be accepted [24, 36, 57]. Therefore, we conclude the convergent validity of the set of constructs reported in Table 5.

Divergent validity was validated using inter-construct correlations, as reported in Table 6. Quadratic correlations between every two constructs are all smaller than the correspondent AVE value shown on the diagonal, which represents an adequate level of discriminant validity [95].

In a preliminary validation, in which we considered the transparency construct understanding input/output as two independent constructs (understanding input *and* understanding output, as proposed in Reference [43]), the analysis of discriminant validity indicated that the two constructs should not be regarded independently, since the quadratic correlation between Input and Output (0.54) was higher than the AVE of Input (0.438) and Output (0.414). Therefore, our subsequent CFA and SEM analyses were run considering the composite construct of understanding Input/Output.

A similar situation arose when analysing the constructs trusting intentions and effectiveness. Here, we found that the quadratic correlation between these two constructs (0.607) was higher than the AVE of trusting intentions (0.481), so we could not conclude a discriminant validity between the two constructs. Therefore, we decided to conduct the subsequent SEM without the construct trusting intentions, while keeping the Trusting Beliefs construct, with which we found discriminant validity with respect to Effectiveness (see Table 6).

7.2.2 SEM Results. As a next step, we ran a first SEM considering all constructs resulting from the convergent and divergent validity analysis (as listed in Table 5). Although the model shows a good fit of the data (*CFI* = 0.948, *TLI* = 0.943, *RMSEA* = 0.039, *SRMR* = 0.066), we found no significant evidence that intuitive decision-making nor visualization familiarity had an influence on any constructs of interest. For example, no significant direct effects were found on perceived explanation quality of: intuitive decision-making (s.p.w. of 0.06, $p = 0.34$), or visualization familiarity (s.p.w. of 0.02, $p = 0.69$); no significant direct nor indirect effects on perceived transparency-information

Table 5. Test Results of Internal Reliability and Convergent Validity

Constructs	Internal reliability		Convergent validity	
	Cronbach alpha	Factor loading	Composite reliability	Average variance extracted
Perceived Expl. Quality	0.875		0.878	0.582
The explanations make me confident that I would like the hotel I chose. [62]		0.554		
The explanations make the recommendation process clear to me. [62]		0.570		
I would enjoy using a recommender system if it presented explanations in this way. [62]		0.718		
Explanations were convincing. [62]		0.670		
The explanations provided sufficient information to make my decision. [32]		0.659		
Perceived Transparency: Information Provision	0.686		0.686	0.422
The system provided information to understand why the items were recommended. [43]		0.568		
The system provided information about how my preferences were inferred. [43]		0.582		
The system provided information about how well the recommendations match my preferences. [43]		0.580		
Perceived Transparency: Understanding Input / Output	0.829		0.827	0.486
It was clear to me what kind of data the system uses to generate recommendations. [43]		0.579		
I understood what data was used by the system to infer my preferences. [43]		0.630		
I understood why the items were recommended to me. [43]		0.527		
I understood how the quality of the items was determined by the system. [43]		0.531		
I understood why the system determined that the recommended items would suit me. [43]		0.597		
Perceived Transparency: Interaction	0.699		0.724	0.656
I know what actions to perform in the system so that it generates better recommendations. [43]		0.805		
I know what needs to be changed in order to get better recommendations. [43]		0.553		
Perceived Trust: Trusting Beliefs	0.669		0.671	0.503
I believe that the recommender would act in my best interest. [80]		0.503		
Recommender is competent and effective in providing hotel recommendations. [80]		0.551		
Perceived Effectiveness	0.821		0.823	0.616
I would recommend the system to others. [61]		0.702		
I could make better decisions with the help of the system. [61]		0.629		
The recommender is useful. [61]		0.539		
Rational Decision Making Style	0.864		0.863	0.611
I thoroughly evaluate decision alternatives before making a final choice. [41]		0.547		
In decision making, I take time to contemplate the pros/cons or risks/benefits of a situation. [41]		0.543		
Investigating the facts is an important part of my decision-making process. [41]		0.517		
I weigh a number of different factors when making decisions. [41]		0.520		
Intuitive Decision Making Style	0.841		0.853	0.587
When making decisions, I rely mainly on my gut feelings. [41]		0.825		
My initial hunch about decisions is generally what I follow. [41]		0.670		
I make decisions based on intuition. [41]		0.815		
I weigh feelings more than analysis in making decisions. [41]		0.503		
Visualization Familiarity	0.862		0.872	0.592
I am competent when it comes to graphing and tabulating data. [62]		0.665		
I frequently tabulate data with computer software. [62]		0.999		
I have graphed a lot of data in the past. [62]		1.076		
I frequently analyze data visualizations. [62]		0.894		

Cronbach's alpha >0.5 are acceptable [93]. Factor loadings >0.5 are acceptable [93]. Average variance extracted (AVE) >0.5 are commonly accepted [36]. However, constructs with AVE values >0.4 whose composite reliability >0.6 can still be accepted [24, 36, 57].

Table 6. Inter-construct Correlation Matrix

	1	2	3	4	5	6	7	8	9
1. Perceived Expl. Quality	**0.582**	0.368	0.192	0.093	0.424	0.490	0.035	0.003	0.005
2. Perceived Transparency: Information Provision	0.607	**0.422**	0.324	0.116	0.270	0.234	0.024	0.003	0.000
3. Perceived Transparency: Understanding Input/Output	0.438	0.569	**0.486**	0.162	0.241	0.174	0.040	0.002	0.007
4. Perceived Transparency: Interaction	0.305	0.341	0.403	**0.656**	0.083	0.071	0.014	0.008	0.037
5. Perceived Trust: Trusting Beliefs	0.651	0.520	0.491	0.288	**0.503**	0.442	0.021	0.002	0.002
6. Perceived Effectiveness	0.700	0.484	0.417	0.267	0.665	**0.616**	0.014	0.004	0.000
7. Rational Decision Making Style	0.188	0.154	0.199	0.120	0.144	0.119	**0.611**	0.045	0.018
8. Intuitive Decision Making Style	0.050	0.053	0.049	0.087	0.040	0.063	−0.212	**0.587**	0.024
9. Visualization Familiarity	0.071	−0.022	0.086	0.193	0.042	0.016	0.135	−0.156	**0.592**

Average Variance Extracted (AVE) on the main diagonal. Correlations below the diagonal. Quadratic correlations above the diagonal. Quadratic correlations between every two constructs are all smaller than the correspondent AVE value shown on the diagonal, which represents an adequate level of discriminant validity [95].

provision of: intuitive decision-making (s.p.w. of 0.06, p = 0.34), or visualization familiarity (s.p.w. of 0.02, p = 0.69). Furthermore, we found no significant moderation effect of visualization familiarity between interface type and explanation quality (B = 0.01, p = 0.53), nor of visualization familiarity between interactivity degree and explanation quality (B = 0.01, p = 0.51). Also, we found no significant moderation effect of intuitive style between interface type and explanation quality (B = 0.01, p = 0.45), nor of intuitive style between interface type and explanation quality (B = 0.02, p = 0.38).

Therefore, we decided to run a subsequent and final SEM omitting the variables: intuitive decision-making and visualization familiarity. The resultant model shows a good fit for the data (*CFI* = .965, *TLI* = 0.961, *RMSEA* = 0.036, *SRMR* = 0.052), a better fit compared to the initial SEM described above. Results are depicted in Figure 13.

Direct effects. The significant positive direct effect of interactivity degree on perceived explanation quality (see Figure 13) suggests that allowing users to dig into the details that support aggregations on positive and negative items' aspects, has a positive impact on users' evaluation of the quality of the explanations. Perceived explanation quality is additionally influenced by rational decision making, so that more rational users reported higher scores of perceived explanation quality.

Furthermore, the significant direct effect from interface type on perceived explanation quality (see Figure 13) indicates that users perceived explanations provided through GUI navigation as better in terms of explanation quality, compared to explanations provided as natural language conversation (see also detailed comparison of means in Table 7).

In regard to perceived transparency, we found a direct effect of perceived explanation quality on information provision, so that the perception of system functions that explicitly inform the way in which recommendations are calculated is influenced by explanation quality (higher scores of perceived explanation quality lead to higher scores of perception of transparency, in regards of information provision). We also found a direct effect of information provision on understanding input and output, and from understanding input and output to interaction. The above path suggests that a greater perception of information provision (through higher quality explanations) leads to a better understanding of the input used by the system and understanding of how or why those recommendations are inferred, leading to a better understanding of what needs to be done by the user, to get better recommendations.

Concerning to perceived trust, we found a direct effect of perceived transparency (regarding information provision, understanding of output and interaction) on trusting beliefs, suggesting that a greater system transparency leads users to a greater perception that the system is honest, and that they can trust its recommendations.

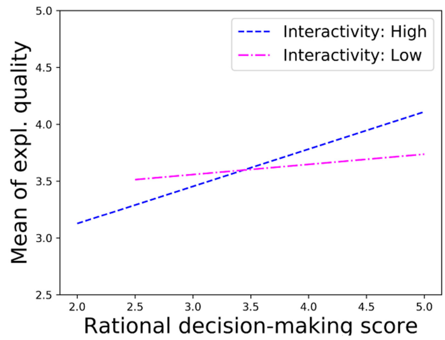

Fig. 12. Interaction plot (fitted means of individual scores) between interactivity degree and rational decision-making style on explanation quality.

In regard to perceived effectiveness, we observed a direct effect of perceived explanation quality and trusting beliefs on it, suggesting that greater explanation quality and greater trusting beliefs lead users to be more confident that they can make better decisions by using the recommender.

Indirect effects. Although a direct effect of interactivity degree on transparency constructs was not found, we observed that explanation quality mediates the effect of interactivity degree on information provision [interactivity degree → perceived explanation quality → information provision], standardized parameter weight of 0.06, $p < 0.05$.

Furthermore, despite finding no direct effect of explanation quality on constructs of transparency: understanding input/output, and interaction, we found an indirect effect of explanation quality on such variables, considering the paths: [explanation quality → information provision → understanding input/output], with a standardized parameter weight of 0.16, $p < .01$; and [explanation quality → information provision → understanding input/output → interaction], with a standardized parameter weight of 0.03, $p < 0.05$.

Moderation and interaction effects. We found a significant moderation effect of rational decision-making between interactivity degree and explanation quality (B = 0.03, $p < 0.05$).

Here, for users in the high interactivity condition, an increase in the rational decision-making score was associated with an increase in their assessment of explanation quality, as depicted in Figure 12. However, we found no significant moderation effect of rational decision-making between interface type and explanation quality (B = 0.02, $p = 0.09$).

Also, we found no significant interaction effect between interactivity degree and interface type on explanation quality (s.p.w. of 0.02, $p = 0.08$).

Although the interactions between rationality and interface type, as well as between interactivity and interface type were not found to be significant, considering that p-values were below 0.1, we do not exclude the possibility of such interactions; we cannot, however, confirm them in this study.

Means of every construct under the combination of the two factors (interactivity degree and interface type) are reported in Table 8.

7.2.3 *Qualitative Evaluation.*

Comments and suggestions by participants. We analyzed the answers of participants to the open ended question, asking for opinions about the system and its explanations, and categorized the responses of participants (Figure 14). Here, we follow an inductive category formation [79]. Based

Table 7. Mean Values and Standard Deviations of Evaluation of RS, Aggregated per Interface Type, and Interactivity Degree (n = 223)

Construct	Interactivity degree				Interface Type			
	High		Low		Natural Lang.		GUI Nav.	
	M	SD	M	SD	M	SD	M	SD
Perceived Expl. Quality	3.89	0.60	3.67	0.73	3.69	0.76	3.87	0.57
Perceived Transparency								
- Information Provision	3.71	0.60	3.49	0.75	3.53	0.75	3.67	0.63
- Understanding Input/Output	3.80	0.58	3.59	0.68	3.66	0.65	3.72	0.62
- Interaction	3.48	0.79	3.15	0.77	3.37	0.82	3.26	0.77
Perceived Trust								
- Trusting Beliefs	3.84	0.60	3.74	0.67	3.77	0.71	3.81	0.57
Perceived Effectiveness	3.99	0.65	3.82	0.72	3.86	0.78	3.96	0.58

Values reported with a 5-Likert scale; higher mean values correspond to a positive evaluation of the RS.

Table 8. Mean Values and Standard Deviations of Evaluation of RS, Per Experimental Condition, Reflecting Combination of Factors: Interface Type and Interactivity Degree (n = 223)

Interactivity degree	High				Low			
Interface Type	Natural Lang.		GUI Nav.		Natural Lang.		GUI Nav.	
Construct	M	SD	M	SD	M	SD	M	SD
Perceived Expl. Quality	3.83	0.67	3.95	0.53	3.55	0.82	3.78	0.61
Perceived Transparency								
—Information Provision	3.63	0.63	3.79	0.57	3.44	0.84	3.54	0.66
—Understanding Input/Output	3.75	0.54	3.84	0.61	3.57	0.75	3.60	0.62
—Interaction	3.51	0.81	3.45	0.78	3.24	0.82	3.07	0.72
Perceived Trust								
—Trusting Beliefs	3.87	0.64	3.82	0.57	3.67	0.76	3.80	0.56
Perceived Effectiveness	3.97	0.73	4.02	0.56	3.74	0.83	3.90	0.59

Values reported with a 5-Likert scale; higher mean values correspond to a positive evaluation of the RS.

on the resulting list of categories and the developed annotation guidelines, the list of user-reported statements was annotated by three members of the research team, who came to an agreement, who came to a Fleiss' kappa = 0.63, a substantial agreement intercoder reliability [65].

Positive comments regarding condition natural language conversation—high degree of interactivity referred to the flexibility of questions that could be asked, and the option to filter comments by polarity (e.g., "I felt the explanations provided were good, because they were drawn on reviews. I liked that reviews could be filters between positive and negative and I could ask comparitive (sic) questions between different hotel options"; "I tried 3–4 questions and it highlighted most of the things I was looking for. The only one it did not provide answers for was whxih (sic) hotel is closest to the beach but overall I liked it"). Further positive comments to the natural language conversation condition pointed to conciseness (e.g., "The explanation (sic) were very simple which is good as it sticks to the facts, Sometimes reviews can get off point and ramble on I liked that you didn't get bogged down with too much information especially is you are comparing a few hotels") and clarity ("the explanations were clear and easy to understand, quick to read"),

Positive comments under the GUI navigation condition with a high degree of interactivity also reported on the benefit of showing overall ratings and their connection to comments extracted

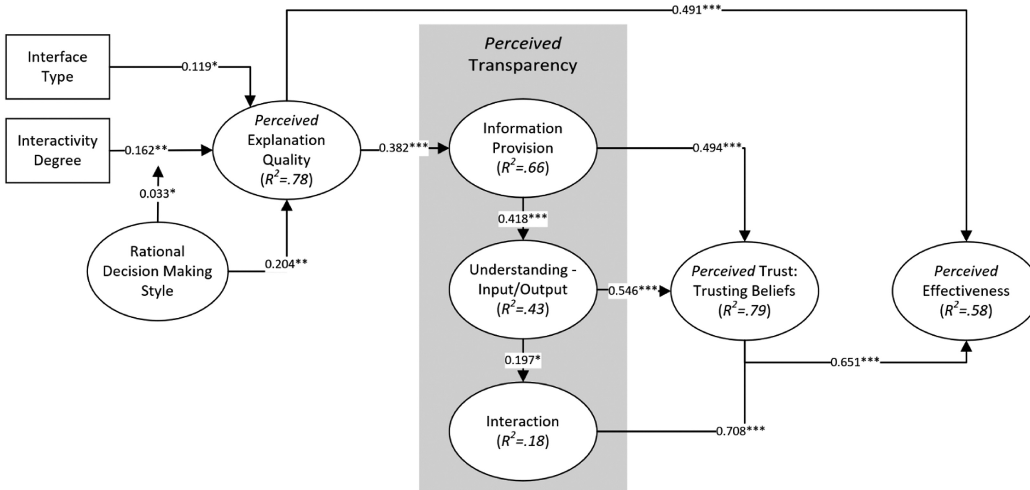

Fig. 13. Structural equation model analysing the influence of degree of interactivity and interface type. The edges show standardized parameter weights (*p < 0.05, **p < 0.01, ***p < 0.001). The amount of explained variance for endogenous variables is displayed inside the nodes.

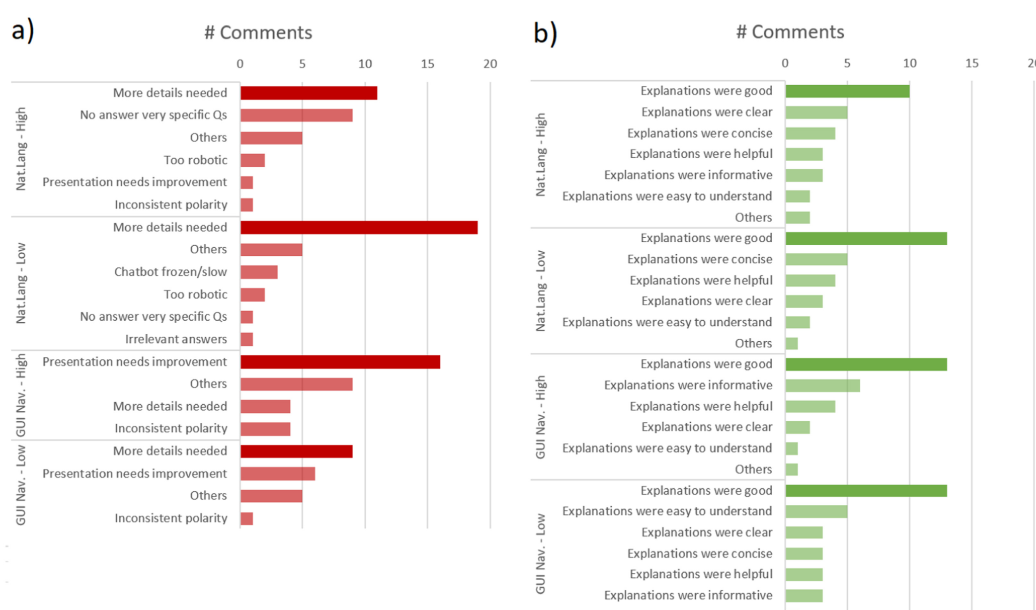

Fig. 14. Type of comments and suggestions by participants, about the system and its explanations, per condition: (a) negative comments (light red for less frequent comments) and (b) positive comments (light green for less frequent comments).

from reviews (e.g., "I enjoyed the fact that I was shown overall ratings for each aspect of the hotel and also had the option to look further into the reviews for each category"). However, users reported on the lack of further information supporting the aggregations under low interactivity degree (e.g., "the explanations were just based on the the percentage of people who liked certain characteristics, so I would need to know more about why they were rated highly by taking the full text into account. Also, it wasn't quite clear how a hotel could get a five star rating, when one aspect only had 69% positive, so the rating system could be clearer"; "I think the explanations provided were clear and concise. It would be nice if quotes from reviews were provided on the aspect I asked about, so that I could see what real people are saying without having to dig through reviews").

In regard to explanations as natural language conversation, comments regarding not answering very specific questions involve remarks such as: "there are some holes in its knowledge, it needs to know room sizes, it's ok"; "The explanations provided were useful, in terms of quickly filtering to see hotels highlighted with specific amenities/features. However, not all questions could be answered so it would be more useful if the system had a bigger bank of information about the hotels"; "I asked a couple of questions about hotels that had early or late book in options which it didn't have any info on..." These types of comments are in line with findings reported in Section 6.

Also, we found that under the GUI navigation—high degree of interactivity condition, the most frequent negative comment related to presentation aspects, e.g., "I disliked having to go back so often to compare hotels. I wish the information could be loaded on the page with all the other recommended options and you could dismiss options as you go"; "In general the explanations were clear but you do have to dig down a lot to get to the detail. I think i the list of hotels displayed the ratings on the front oage (sic) and maybe allowed you to sort based on your preferences it would be even easier to use."

Type of information input used for recommendations. We analyzed the answers of participants to the open ended question, asking for how they would explain to a friend how the system generated the recommendations. We categorized the responses of participants, according to the type of information related in participants' comments (Figure 15). Here, we also follow an inductive category formation [79]. Based on the resulting list of categories and the developed annotation guidelines, the list of user-reported statements was annotated by three members of the research team, who came to a Fleiss' kappa = 0.67, a substantial agreement intercoder reliability [65].

7.3 Discussion: Evaluation of GUI Navigation and Natural Language Conversation Interfaces

7.3.1 Quantitative Evaluation.

Effects of interactivity degree. Our results show that higher degree of interactivity has a significantly positive effect on users' evaluation of explanation quality, independent of the type of interface, compared to explanations with a lower degree of interactivity, thus confirming our **H1**. The above in turn confirms the suitability of our proposal for explanations as interactive argumentation, inspired by dialog models of explanation, which enables users to question initial explanation attempts provided by the system, in particular the aggregate representation of positive and negative customer opinions. Here, active control and two-way communication as addressed in our proposal (in both natural language and GUI navigation interfaces) seem to play a role in the observed effect, namely, active control, as users can exert control over the argumentative content to be displayed by the system; two-way communication, as users can inform the system which statements require further backing, and which features are of momentary relevance to them, thus contributing to a better understanding of reasons behind recommendations, as sustained by dialog models of explanation [51, 114], and as revealed by our analysis: a higher perceived explanation quality significantly leads to a more positive perception of system functions that inform how recommendations are generated. The above in turn is related to a greater understanding of the input information, of how recommendations are matched to user preferences, and what actions can be taken to receive better recommendations. In turn, we found that this understanding significantly

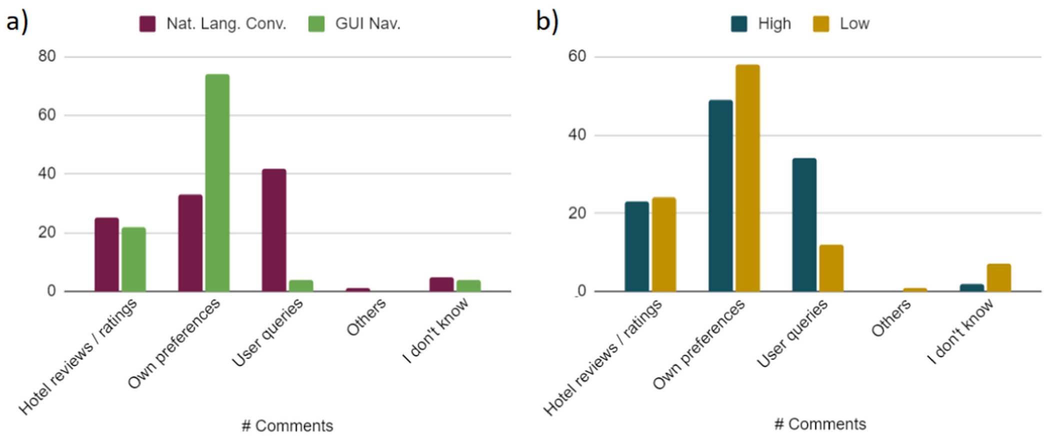

Fig. 15. Type of information input used by the system to generate recommendations, according to participants: (a) by interface type and (b) by interaction degree.

leads to higher scores of perceived trust and effectiveness, confirming thus our **H2**, **H3** and **H4**. As we also observed a (small) moderating effect of the user's degree of rationality on the relation between interactivity and perceived explanation quality, it seems , however, that personal characteristics, and possibly other contextual factors not addressed in this study, may have an impact on the suitability of high-interactivity explanations.

The above aligns with the causal relationships among RS evaluation constructs found by Pu et al. [95], as well as with the evaluation framework proposed by Knijnenburg et al. [61]: objective system aspects (different interactivity degrees in our case) influence subjective system aspects (e.g., perceived explanation quality), which in turn influence users' attitudes and behaviours (e.g., trusting beliefs and intentions), and users' experience (e.g., perceived effectiveness).

Furthermore, our results suggest that questionnaire items to evaluate perceived transparency proposed by Reference [43] appear to be a significant means of addressing transparency in RS more comprehensively, allowing testing of the impact of explanation features on particular aspects of transparency. This could be investigated in our analysis in more detail than was possible with previous measurement instruments, which address transparency as the general understanding of why items are recommended, as in Reference [95].

Effects of interface type. Providing interactive explanations in RS under our proposed approach, and using both types of interface (natural language and GUI navigation), proved to improve users' assessment of the system, given the predominant positive evaluation of the two tested systems and their explanations by the participants. Although we found a significant effect of interface type on on explanation quality, we could not confirm our **H5** (users' evaluation of transparency is more positive when they interact with an explanatory natural language interface). Instead, the results suggest a greater preference for explanations provided through GUI navigation over those provided through natural language conversation.

In this regard, we acknowledge that direct comparison of two such different interfaces is challenging, there may be many underlying differences not captured by the study design or the questionnaires, making it difficult to draw conclusions, although the two interface designs share the same information flow, according the proposed explanatory structure (Section 3). Nevertheless, we discuss below design features to consider when designing explanatory interfaces under the two interface types and their possible trade-offs.

The natural language interface may entail advantages for certain users. Under this approach, however, the user must take the initiative to formulate their own questions, which competes with a more proactive position on the part of the system in the case of GUI navigation, where it provides a more comprehensive summary of opinions (covering a greater number of aspects at the same time in a table), without expecting the user to inquire about specific aspects. Additionally, visual arguments (a combination of visual and verbal information, e.g., using a table as in the GUI navigation) may have a greater "rhetorical power potential" than verbal arguments, due (among others) to their greater immediacy (possibility of quick processing) [9].

The display of a global, graphical evaluation of customer opinions, referencing a wider range of aspects in a single table (GUI Navigation) contrasts to the natural language approach, in which aggregations of opinions were indicated only up to two aspects (following the principle of brevity suggested by Reference [85], answers should be provided as concisely as possible). The natural language approach focuses on answering specific user questions, but at the same time involves a shortcoming: it makes it more difficult to users to have a look at aspects that may not be of their greatest interest, but may be decisive in making a final decision, especially if they have customer reports that were not so favorable.

One might reasonably assume that using a graphical user interface may result in shorter times to access the required information. However, we did not draw conclusions regarding interaction times, since in our crowdsourced study many variables beyond our control could affect the measurements. Future lab studies will be needed to gain insights into the detailed interaction behavior. Additionally, explaining recommendations by means of GUI navigation implies an advantage over the natural language interface, since, as in the case of direct-manipulation style interfaces [101], they can provide "consistent and concrete representations of data, operations, and system states" [11], as well as clear options suggesting what the user can do next. However, using natural language interfaces might increase the users' expectations in regard to what a conversational agent can actually perform [54]. The above can, however, be mitigated by a continuing use of the application, as well as presenting new users with examples on how to interact with the conversational agent, as well as the type of questions the user can ask [54], guidelines we followed in our user study. The above contributed, in our opinion, to the predominantly positive evaluation we observed in the natural language condition.

Influence of user characteristics. In regard to the effect of user characteristics, the use and perceived benefit of interaction options might be influenced by individual differences, as discussed in Reference [71]. While we found no significant evidence that intuitive decision-making style nor visualization familiarity have a significant moderation effect of interface type on explanation quality (**H6**), we did find that the way people process information when making decisions play a significant role in the evaluation of explanations as natural language conversation, as also reported in Reference [50] for GUI navigation.

In particular, we found that the high interactivity condition had a more positive effect on the perception of explanation quality the higher participants' rationality score was, thus confirming a moderating effect of rationality, which was only observed for high interactivity

(**H7**). The above could be explained by the propensity of people with a predominant rational decision-making style, to search for information and evaluate alternatives exhaustively [41], which is facilitated by our approach. This suggests that interactive review-based explanations seem to benefit more the users who tend to evaluate information thoroughly when making decisions, compared to users who use a more shallow information-seeking process.

7.3.2 Qualitative Evaluation. When analyzing the comments reported by users regarding their perception and suggestions for improving the system, we found that most of the negative comments reported by participants refer to the lack of additional details: (1) under the low interactivity condition, a lack of access to customer comments supporting the explanations, and (2) under the natural language interface condition, the lack of specific details in answers to factoid questions (such as, for example, the type of menu offered in the hotel's restaurant, or distance to the beach).

In relation to the open-ended question, in which participants were asked to indicate how they would explain to a someone else how the system generates recommendations, we noted an overall fair understanding of what kind of information the system uses to generate recommendations, that is, the participants' responses mostly reflected the input used, rather than specific details about the algorithm, for which our explanations do not provide details. On the one hand, the number of responses mentioning that the system uses hotel reviews and ratings is very similar across the conditions. On the other hand, a salient difference can be observed in comments referring to preferences as a base for recommendations, in particular between different interface types. Here, participants under the GUI navigation condition reported much more comments indicating that recommendations were based on their own preferences.

The above seems to be the effect of showing explanations using a list view of preferred aspects and aggregated customer opinions, within the same table, while in the natural language case, the responses indicating that recommendations are based on preferences are only provided for questions that do not address an aspect in particular (around 12% of asked questions), e.g., Q: "Why does Hotel Amelia have the highest rating?" A: "Because the most important aspects for you were commented positively, 90% about room and 69% about location." However, the lack of awareness of preferences used as input of recommendations might be mitigated by users being aware that recommendations are generated based on their queries. That is, in the natural language condition, the understanding that recommendations can be refined as a result of the conversation seems fostered. The above seems even more pronounced under the natural language condition with a high degree of interactivity, where in addition to a simple exchange of textual questions and answers, additional options are offered, for example, to refine the recommendations given a certain aspect or feature. This qualitative finding is also consistent with the significant difference observed in the scores reported for the *interaction* sub-construct of the transparency scale, i.e., the user can better identify what actions to take and what to change to obtain better recommendations, in the case of explanations with a high degree of interactivity.

Finally, only a low number of comments indicated that they did not know how the system generated the recommendations (nine comments in total, seven under low degree of interactivity), or indicated reasons that had nothing to do with the actual functioning of the system (only one comment related to using preference information from third parties), which suggests that our interactive approach to explanations contributes to a great extent to the understanding of the information that the system uses to generate its recommendations.

8 CONCLUSIONS AND OUTLOOK

We discussed in Section 3 a scheme for explanations as interactive argumentation in review-based RS, inspired by dialog explanation models and formal argument schemes, that allows users to go from aggregated customer opinions to detailed extracts of individual reviews, to facilitate a better understanding of the claims made by the RS. We found that providing interactive explanations in RS based on the proposed scheme proved to be an effective means for improving users' evaluation of the system, both in the GUI navigation and natural language conversation variants of the RS implemented.

We also found that providing a higher degree of interactivity in explanations contributed to a more positive evaluation of transparency, effectiveness, and trust in the system, independent of the interactive interface type. We furthermore found that individual differences in terms of user characteristics (e.g., decision-making style) may lead to differences in the evaluation of the proposed implementation.

Based on our findings concerning language-based conversational explanations, we conclude that our proposal of a dimension-based intention model is a valid approach to represent user queries in the context of explanatory RS. By providing ConvEx-DS, a data set annotated based on this model, we also hope to contribute to the development of future dialog systems that support conversational explanations in RS.

8.1 Limitations and Future Work

Despite our motivating results, it is important to note the limitations imposed by the discussed proposal. First, our approach involves providing excerpts of customer reviews as backing for aggregation-type explanations, filtered by aspect and sentiment. However, we did not implement methods for sorting customer comments according to their helpfulness or argumentative quality. Thus, we plan in the future to integrate techniques that will allow us to take the quality of comments into account, as well as to evaluate their effect, both on the accuracy of recommendation prediction and on the evaluation of the system by end-users.

Furthermore, our proposal has been specifically applied to the hotel domain. However, we suppose that our approach will generalize properly to domains related to experience goods [88], where the search for information during decision making relies heavily on word-of-mouth [60, 89], as is the case, for example, for restaurants. Thus, in future work, we will evaluate the use of our explanation scheme in domains other than hotel, as well as the validity of our intention model, and the usefulness of ConvEx-DS.

In regard to our conversational approach, addressing intent detection as a text classification problem by means of an intent classification model allows to provide answers that approximate the information need expressed by the user. However, the approach is insufficient when dealing with questions that are too specific, particularly in regard to factoid questions. Consequently, the development of a conversational agent with explanatory purposes in RS should not only rely on the underlying RS algorithm, customer reviews or hotels metadata (as in our developed system), but should also integrate further sources of information, e.g., external location services, to provide specific details, such as surroundings, distances to places of interest or transport means, in case these are not found in customer reviews or metadata.

Furthermore, as reported in Section 7, participants under the natural language condition acknowledged that the system refined the recommendations as a result of the conversation with the system. However, the system was not designed as a critiquing RS as such, where users can directly express their preferences by modifying features of recommended items. In particular, the dialog policy designed as the basis of the ConvEx system focused on the support of the explanation process, rather than on the elicitation of preferences. Therefore, we plan in the future to evaluate and implement a system that blends the functionality of critiquing systems with that of conversational explanations to improve user experience.

Finally, conversations in ConvEx are based on a dialog manager that falls into the handcrafted category of dialog management approaches, where the state and policy are defined as a set of rules defined by developers and domain experts. While suitable for our proposed dialog policy, and for testing our hypothesis in Section 7, more flexible approaches would be needed to extend the policy for supporting preference elicitation as well as a broader range of user goals such as item search, customer service or booking. End-to-end neural network approaches seem promising for this purpose, as they allow for a more dynamic modeling of possible dialogues with different goals (as proposed by Reference [72]), an alternative that we consider exploring in the future.

APPENDICES

A CONVEX-DS COLLECTION AND ANNOTATION ARTIFACTS

A.1 WoOz Pre-study

We used the following scheme (Figure 16) as the guideline for the wizard (Section 6.1), aiming to portray a structured conversation similar across participants.

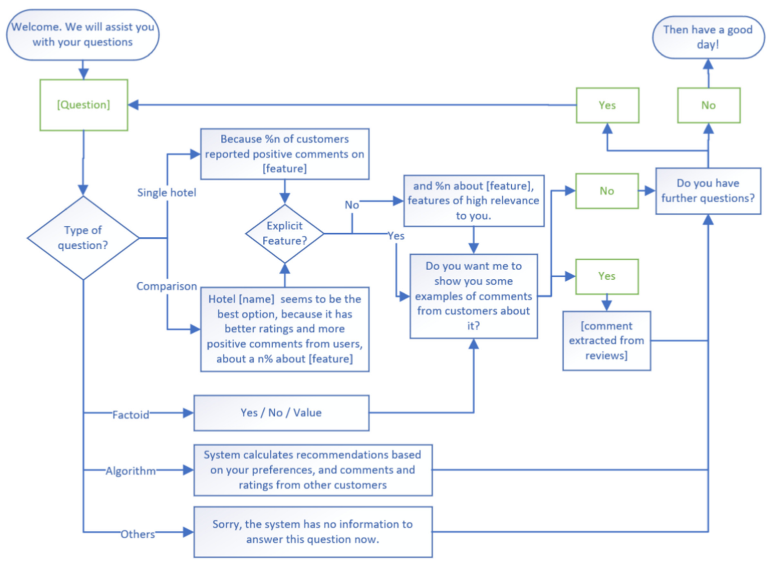

Fig. 16. Wizard guideline for conversational explanations used in WoOz experiment (Section 6.1), as reported in Reference [48]. Blue boxes represent utterances by the system, green boxes represent the utterances by users.

A.2 Corpus Collection: Annotation Guidelines

Figures 17, 18, and 19 depict the guidelines presented to the annotators for the process of annotating the corpus described in Section 6.2.

Please read each question and choose the correct category. Click on "Next question" to continue with the next question. Once you reach the end (100 questions), come back to the main survey, and click on the button "Next", to obtain the Completion Code to copy and paste into Prolific.

Classification - Instructions reminder:

Comparative: if the question expresses a relation based on similarities or differences of more than one hotel or object. Includes expressions like "greater", "more than", "less than", "better", "worst", "best".

Examples:

"Which is the cheapest hotel?" *"why does hotel levi cost less?"* *"which hotel is closest to the city center?"* *"Which hotel was better for location purposes?"* *"Which has the best facilities?"*

Non-comparative: Does not expresses a comparison.

Examples:

"What facilities does Hotel Henry have?" *"Which hotels have bathtubs in?"* *"Is Hotel Henry for business people?"* *"why is the hotel Emily so expensive?"* *"is this hotel good?"*

Fig. 17. Annotation guideline, dimension comparison.

Classification - Instructions reminder:

Factual: Refers to the existence or characteristics of hotel features (facts). This questions could be answered, for example, with a Yes / No answer, check in times, list of facilities, or price of a room or breakfast.

Examples:

"does hotel Penolpe have wifi?"
"what time is check in for Hotel Evelyn?"
"what facilities are available at Hotel Henry?"
"What is the price of a single room?"

Subjective: Refers to a subjective assessment of the hotel or its characteristics. Includes expressions like "how good", "how close / far", "how expensive / cheap", "which is the best", "is it nice".

Examples:

"which hotel has the best location"
"how close is the hotel to main attractions?"
"which is the closest to a station?"
"How comfortable is the room"
"does hotel Hannah have nice food?"
"how expensive is this hotel?"

Why recommended: Question about system reasons to recommend/ not recommend a hotel, or **why** a hotel (or feature) is good / bad, or **why** is it better / worst than others.

Examples:

"Why is Hotel Emily in a good location?"
"Why did you recommend Hotel Emily?"
"Why is the price at the Hotel Amelia good?"
"Why does Hotel Hannah have the highest rating?"

Fig. 18. Annotation guideline, dimension assessment.

Classification - Instructions reminder:

Aspect: Question inquires for an specific aspect or feature of the hotels.

Examples:

"Are these the lowest priced hotels?" (aspect here: price)
"Why are Hotel Joseph facilities so good?" (aspect here: facilities)
"which room is the biggest?" (aspect here: room)
"Which hotels have good service?" (aspect here: staff)

Overall: Question doesn't address any hotel aspect or feature in particular.

Examples:

"Why did you pick hotel Penelope?"
"What are the problems with hotel Emily?"
"Why is Hotel Emily a good choice?"
"Which hotel has the most positive reviews?"

Fig. 19. Annotation guideline, dimension detail.

B USER EVALUATION QUESTIONNAIRES

Table 9 shows the questionnaire items used in the user study reported in Section 7.

Table 9. Item Questions Used in User Study Reported in Section 7

Code	Question	Analisys Name Variable	Reference
UT02_01	I prefer to gather all the necessary information before committing to a decision.	UT_rational	[41]
UT02_02	I thoroughly evaluate decision alternatives before making a final choice.	UT_rational	[41]
UT02_03	In decision making, I take time to contemplate the pros/cons or risks/benefits of a situation.	UT_rational	[41]
UT02_04	Investigating the facts is an important part of my decision-making process.	UT_rational	[41]
UT02_05	I weigh a number of different factors when making decisions.	UT_rational	[41]
UT02_06	When making decisions, I rely mainly on my gut feelings.	UT_intuitive	[41]
UT02_07	My initial hunch about decisions is generally what I follow.	UT_intuitive	[41]
UT02_08	I make decisions based on intuition.	UT_intuitive	[41]
UT02_09	I rely on my first impressions when making decisions.	UT_intuitive	[41]
UT02_10	I weigh feelings more than analysis in making decisions.	UT_intuitive	[41]
UT02_18	I am competent when it comes to graphing and tabulating data.	UT_visual_fam	[62]
UT02_19	I frequently tabulate data with computer software.	UT_visual_fam	[62]
UT02_20	I have graphed a lot of data in the past.	UT_visual_fam	[62]
UT02_21	I frequently analyze data visualizations.	UT_visual_fam	[62]
EE02_06	I would recommend the system to others.	effectiveness	[61]
EE02_07	I could make better decisions with the help of the system.	effectiveness	[61]
EE02_08	The recommender is useful.	effectiveness	[61]
EE02_10	I believe that the recommender would act in my best interest.	trusting_beliefs	[80]
EE02_11	I would characterize recommender as honest.	trusting_beliefs	[80]
EE02_12	Recommender is competent and effective in providing hotel recommendations.	trusting_beliefs	[80]
EE02_13	I would feel comfortable depending on the information provided by recommender.	trusting_intentions	[80]
EE02_15	I would confidently book a hotel based on recommendation I was given by recommender.	trusting_intentions	[80]
EE02_16	I would be willing to share the specifics of my preferences when looking for a hotel to the recommender.	trusting_intentions	[80]
EE05_04	The explanations make me confident that I would like the hotel I chose.	explanation_quality	[62]
EE05_05	The explanations make the recommendation process clear to me.	explanation_quality	[62]
EE05_06	I would enjoy using a recommender system if it presented explanations in this way.	explanation_quality	[62]
EE05_08	Explanations were convincing.	explanation_quality	[62]
EE05_09	Explanations were easy to understand	explanation_quality	[32]
EE05_20	The explanations provided sufficient information to make my decision.	explanation_quality	[32]
EE03_01	Please let us know about your overall opinion about the explanations provided or how they could be improved:	comments	[47]
EE02_27	The system provided information to understand why the items were recommended.	transp_info_provision	[43]
EE02_28	The system provided information about how the quality of the items was determined.	transp_info_provision	[43]
EE02_29	The system provided information about how my preferences were inferred.	transp_info_provision	[43]
EE02_30	The system provided information about how well the recommendations match my preferences.	transp_info_provision	[43]
EE02_22	It was clear to me what kind of data the system uses to generate recommendations.	transp_input	[43]
EE02_23	I understood what data was used by the system to infer my preferences.	transp_input	[43]
EE02_24	I understood which item characteristics were considered to generate recommendations.	transp_input	[43]
EE02_25	I understood why the items were recommended to me.	transp_output	[43]
EE02_31	I understood how the quality of the items was determined by the system.	transp_output	[43]
EE02_32	I understood why the system determined that the recommended items would suit me.	transp_output	[43]
EE02_26	I can tell how well the recommendations match my preferences.	transp_output	[43]
EE02_33	I know what actions to perform in the system so that it generates better recommendations.	transp_interaction	[43]
EE02_34	I know what needs to be changed in order to get better recommendations.	transp_interaction	[43]
HC03_01	Please provide the reasons why you chose the hotel you did.	reasons	
EE07_01	How would you explain to a friend, in your own words, how the system generates recommendations?	explain_to_someone	

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